

Supporting information for

Biaryl Construction through Kumada Coupling with Diaryl Sulfates as One-by-one Electrophiles under Mild Condition

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Table	Page
General	2
Synthesis and analytical data of diaryl sulfates 1	2-5
Synthesis and analytical data of biaryl products 3	6-12
Mechanism exploration	13
¹ H and ¹³ C NMR spectral data	14-45
Reference	46

General: All reactions were carried out under N₂ atmosphere. All the reagents were purchased from Alfa Aesar and Acros Chemical and used without further purification. The Grignard reagents were newly prepared in THF while phenylmagnesium bromide was purchased from Aldrich and used without further purification. The concentration of the Grignard reagents was determined by the Acid-Base Titration. NiCl₂(PCy₃)₂ was synthesized according to literature method.¹ The solvent was freshly distilled over sodium with the use of diphenyl ketone as an indicator. ¹H NMR (300 MHz) and ¹³C NMR (75 MHz) were registered on 300M spectrometers with CDCl₃ as solvent and tetramethylsilane (TMS) as internal standard. Chemical shifts were reported in units (ppm) by assigning TMS resonance in the ¹H spectrum as 0.00 ppm and CDCl₃ resonance in the ¹³C spectrum as 77.0 ppm. All coupling constants (*J* values) were reported in Hertz (Hz). Column chromatography was performed on silica gel 200-300 mesh. GC, HPLC and MS were performed by the Analytical Center in Peking University.

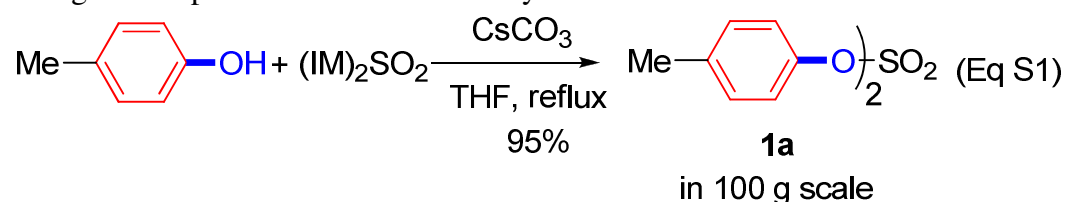
Condition for the Suzuki reaction from compound 18 to compound 19

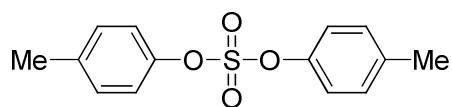
0.2 mmol (80.2 mg) **18**, 0.3 mmol (72.6 mg) phenylboronic acid, 0.02 mmol (10mg) Pd(OAc)₂, 0.4 mmol (84.8 mg) Na₂CO₃ were added in a sealed tube under N₂. 1.4mL H₂O and 1.2 mL acetone were injected in to the mixture. The reaction was stirred for 1 h under room temperature. The reaction was washed by 25mL H₂O and extracted by 25×3mL CH₂Cl₂. The solvent was removed and the pure product was separated through a flash column chromatography (CH₂Cl₂ : PE=1:5) as a white solid. The yield was 84%.

Method A: General procedure for the syntheses of diaryl sulfates:

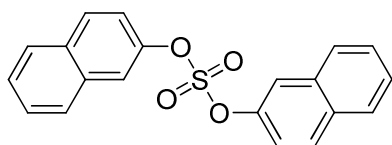
The phenol(15 mmol), 1.63g Cs₂CO₃(5mmol), 0.99g N,N'-sulfuryldiimidazole (5 mmol) and 5mL THF were added in a 20mL of dry tube. Then the mixture was heated to reflux and stirred overnight. The mixture was cooled to room temperature and then filtered through a short silicon column. After concentration of the filtrate, the diaryl sulfates was purified by column chromatography. The yields reported here were based on N,N'-sulfuryldiimidazole and the addition of excess phenol could restrain the generation of monosulfates. The excessive phenol could easily be removed from the system and recycled in large scale.

By a simple scale-up, 100 gram of di(*p*-tolyl) sulfate **1a** was produced without the change of the procedure in an excellent yield.

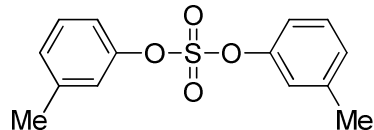




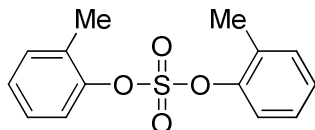
Di(*p*-tolyl) sulfate (1a). The product was made following method A with a yield of 95% as a white solid. ^1H NMR (300 MHz, CDCl_3): δ 7.20 (s, 8 H), 2.36 (s, 6 H). ^{13}C NMR (75.4 MHz, CDCl_3): δ 148.5, 137.6, 130.5, 120.8, 20.8. ESI-MS m/z : 301.0503 $[\text{M}+\text{Na}]^+$, calcd. for $\text{C}_{14}\text{H}_{14}\text{NaO}_4\text{S}$: 301.0510. IR (KBr, cm^{-1}): ν 2924, 1501, 1459, 1378, 1216, 1179, 1145, 881.



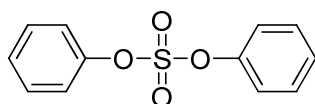
dinaphthalen-2-yl sulfate (1b) The product was made following method A with a yield of 94% as a white solid. ^1H NMR (300 MHz, CDCl_3): δ 7.84-7.93 (m, 8H), 7.54-7.57 (m, 4H), 7.43-7.47 (m, 2H). ^{13}C NMR (75.4 MHz, CDCl_3): 148.0, 133.5, 132.1, 130.4, 128.0, 127.9, 127.3, 126.8, 119.7, 118.7. ESI-MS m/z : 373.0498 $[\text{M}+\text{Na}]^+$, calcd. for $\text{C}_{20}\text{H}_{14}\text{NaO}_4\text{S}$: 373.0510. IR (KBr, cm^{-1}): ν 2925, 1511, 1461, 1415, 1196, 1110, 912.



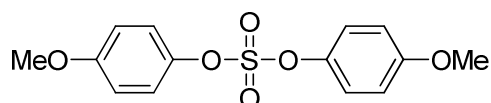
Di(*m*-tolyl) sulfate (1c) The product was made following method A with a yield of 94% as a colorless oil. ^1H NMR (300 MHz, CDCl_3): δ 7.26-7.33 (m, 2 H), 7.11-7.16 (m, 6 H), 2.38 (s, 6 H). ^{13}C NMR (75.4 MHz, CDCl_3): 150.5, 140.6, 129.7, 128.4, 121.6, 118.0, 21.2. ESI-MS m/z : 301.5052 $[\text{M}+\text{Na}]^+$, calcd. for $\text{C}_{14}\text{H}_{14}\text{NaO}_4\text{S}$: 301.0510. IR (KBr, cm^{-1}): ν 2923, 1585, 1488, 1460, 1417, 1201, 1125, 929, 848.



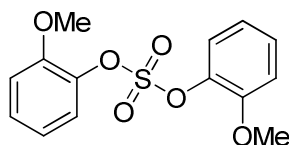
Di(*o*-tolyl) sulfate (1d) The product was made following method A with a yield of 95% as a yellow oil. ^1H NMR (300 MHz, CDCl_3): δ 7.21-7.34 (m, 8 H), 2.32 (s, 6 H). ^{13}C NMR (75.4 MHz, CDCl_3): δ 149.2, 132.0, 131.1, 127.7, 127.4, 121.2, 16.1 MS (EI) m/z : 204 (M^+). ESI-MS m/z : 301.0501 $[\text{M}+\text{Na}]^+$, calcd. for $\text{C}_{14}\text{H}_{14}\text{NaO}_4\text{S}$: 301.0510. IR (KBr, cm^{-1}): ν 2925, 1491, 1398, 1207, 1152, 1097, 882.



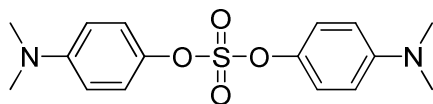
diphenyl sulfate (1e) The product was made following method A with a yield of 90% as a yellow oil. ^1H NMR (300 MHz, CDCl_3): 7.39-7.45 (m, 4H), 7.30-7.36 (m, 6H). ^{13}C NMR (75.4 MHz, CDCl_3): δ 150.5, 130.1, 127.7, 121.1. ESI-MS m/z : 273.0192 $[\text{M}+\text{Na}]^+$, calcd. for $\text{C}_{12}\text{H}_{10}\text{NaO}_4\text{S}$: 273.0197. IR (KBr, cm^{-1}): ν 1487, 1415, 1400, 1215, 1143, 876.



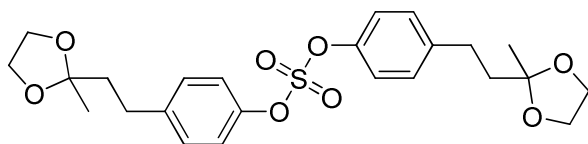
bis(4-methoxyphenyl) sulfate (1f) The product was made following method A with a yield of 89% as a white solid. ^1H NMR (300 MHz, CDCl_3): δ 7.24 (d, $J = 9.0$ Hz, 4 H), 6.91 (d, $J = 9.0$ Hz, 4 H), 3.82 (s, 6 H). ^{13}C NMR (75.4 MHz, CDCl_3): δ 158.7, 144.0, 122.3, 114.9, 55.6. ESI-MS m/z : 333.0403 $[\text{M}+\text{Na}]^+$, calcd. for $\text{C}_{14}\text{H}_{14}\text{NaO}_6\text{S}$: 333.0409. IR (KBr, cm^{-1}): ν 2924, 1500, 1253, 1169, 1032, 884.



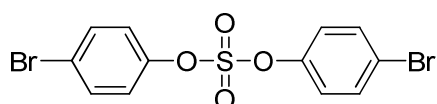
bis(2-methoxyphenyl) sulfate (1g) The product was made following method A with a yield of 95% as a white solid. ^1H NMR (300 MHz, CDCl_3): δ 7.44- (m, 2H), 7.11-7.16 (m, 2H), 7.11-7.16 (m, 4H), 3.87 (s, 6H). ^{13}C NMR (75.4 MHz, CDCl_3): δ 151.7, 139.7, 128.4, 122.8, 120.7, 113.2, 56.0. ESI-MS m/z : 333.0400 $[\text{M}+\text{Na}]^+$, calcd. for $\text{C}_{14}\text{H}_{14}\text{NaO}_6\text{S}$: 333.0409. IR (KBr, cm^{-1}): ν 2924, 1501, 1414, 1283, 1261, 1214, 1154, 1104, 888.



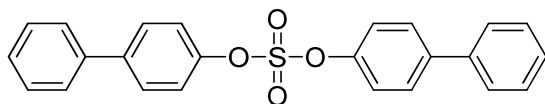
bis(4-(dimethylamino)phenyl) sulfate (1h) The product was made following method A with a yield of 95% as a white solid. ^1H NMR (300 MHz, CDCl_3): 7.17 (d, $J = 9.6$ Hz, 4H), 6.66 (d, $J = 9.3$ Hz, 4H), 2.95 (s, 12H). ^{13}C NMR (75.4 MHz, CDCl_3): δ 149.7, 141.3, 121.8, 112.7, 40.6. ESI-MS m/z : 337.1212 $[\text{M}+\text{Na}]^+$, calcd. for $\text{C}_{16}\text{H}_{21}\text{N}_2\text{O}_4\text{S}$: 337.1222. IR (KBr, cm^{-1}): ν 2921, 1603, 1513, 1446, 1396, 1369, 1235, 1177, 1142, 1068, 948, 873, 813.



bis(4-(2-(2-methyl-1,3-dioxolan-2-yl)ethyl)phenyl) sulfate (15) In a 20mL dry tube, 1.64g 4-(4-hydroxyphenyl)butan-2-one (10 mmol), 1.63g Cs_2CO_3 (5mmol) and 990mg N,N' -sulfuryldiimidazole (5 mmol) were added. Then 5mL THF was injected in the mixture. The reaction was refluxed overnight. The mixture was cooled to room temperature and then filtered through a short silicon column. The solvent was removed and the product (bis(4-(3-oxobutyl)phenyl) sulfate) was purified by column chromatography as a white solid with a yield of 83%. 0.72g bis(4-(3-oxobutyl)phenyl) sulfate (2mmol), 0.37g ethane-1,2-diol (6mmol), 0.084g $\text{TsOH}\cdot\text{H}_2\text{O}$ (0.2mmol) and 10mL toluene were added into a 50mL flask. The mixture was refluxed overnight with a Dean-Stark trap. The reaction was cooled to room temperature and washed by brine (25mL $\times$ 3). The organic layer was dried by the MgSO_4 . The solvent was removed and the product (bis(4-(2-(2-methyl-1,3-dioxolan-2-yl)ethyl)phenyl) sulfate) was purified by column chromatography as a white solid with a yield of 97.5%. ^1H NMR (300 MHz, CDCl_3): δ 7.24-7.27 (m, 8H), 3.96-4.02 (m, 8H), 2.71-2.77 (m, 4H), 1.93-1.99 (m, 4H), 1.37 (s, 6H). ^{13}C NMR (75.4 MHz, CDCl_3): 129.8, 121.0, 109.5, 64.8, 40.8, 29.5, 23.9. ESI-MS m/z : 501.1545 $[\text{M}+\text{Na}]^+$, calcd. for $\text{C}_{24}\text{H}_{30}\text{NaO}_8\text{S}$: 501.1559. IR (KBr, cm^{-1}): ν 2927, 1500, 1413, 1375, 1256, 1212, 1147, 1056, 883.



bis(4-bromophenyl) sulfate (17) The product was made following method A with a yield of 93% as a white solid. ^1H NMR (300 MHz, CDCl_3): 7.55 (d, J = 9.0 Hz, 1H), 7.19 (d, J = 8.7 Hz, 2H). ^{13}C NMR (75.4 MHz, CDCl_3): δ 149.2, 133.3, 122.8, 121.3. ESI-MS m/z : 428.8393 $[\text{M}+\text{Na}]^+$, calcd. for $\text{C}_{12}\text{H}_8\text{Br}_2\text{NaO}_4\text{S}$: 428.8408. IR (KBr, cm^{-1}): ν 1477, 1418, 1218, 1147, 1067, 1012, 878, 831, 775.

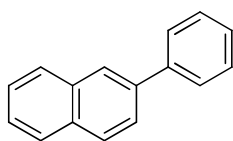


Di(biphenyl-4-yl) sulfate (18) The product was made following method A with a yield of 65% as a white solid. ^1H NMR (300 MHz, CDCl_3): δ 7.63 (d, J = 9.0 Hz, 4H), 7.56 (d, J = 7.2 Hz, 4H), 7.35-7.48 (m, 10H). ^{13}C NMR (75.4 MHz, CDCl_3): δ 149.8, 141.1, 139.7, 129.0, 128.8, 127.9, 127.2, 121.4. ESI-MS m/z : 425.0808 $[\text{M}+\text{Na}]^+$,

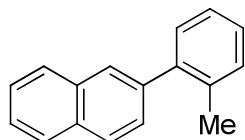
calcd. for $C_{24}H_{18}NaO_4S$: 425.0823. IR (KBr, cm^{-1}): ν 3033, 1510, 1483, 1420, 1400, 1218, 1151, 879, 762.

Method B: General procedure for syntheses of biaryl product 3

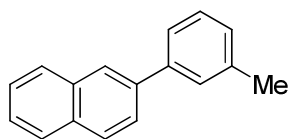
$NiCl_2(PCy_3)_2$ (6.8, 0.01 mmol), PCy_3 (5.6 mg, 0.02 mmol), diarylsulfates (0.2 mmol), newly made Grignard reagent (0.6mL 1M, 0.6mmol) and 2 mL freshly distilled diethylether were added into a dry Schlenk tube under a nitrogen atmosphere at room temperature. The reaction mixture was stirred for 2 h under room temperature. After that, about 0.5g silica gel was added to quench the reaction. Then the mixture was filtered through a short silicon column. After concentration of the filtrate, the biaryl product 3 was purified by column chromatography.



2-phenylnaphthalene.^[1](3ba) The product was made following method B with a yield of 85% as a white solid. 1H NMR (300 MHz, $CDCl_3$): δ 8.00 (s, 1H), 7.79-7.86 (m, 3H), 7.66-7.71 (m, 3H), 7.40-7.45 (m, 4H), 7.30-7.35 (m, 1H). ^{13}C NMR (75.4 MHz, $CDCl_3$): δ 141.2, 138.6, 133.7, 132.7, 128.9, 128.5, 128.2, 127.7, 127.5, 127.4, 126.3, 125.9, 125.8, 125.6. MS (EI) m/z : 204 (M^+). IR (KBr, cm^{-1}): ν 3056, 1597, 1495, 1453, 1265, 894, 859, 821, 738.

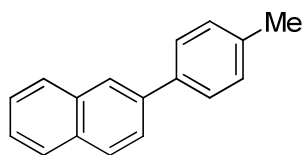


2-(o-tolyl)naphthalene.^[1] (3bb) The product was made following method B with a yield of 93% as a colorless oil. 1H NMR (300 MHz, $CDCl_3$): δ 7.79-7.83 (m, 3H), 7.74 (s, 1H), 7.42-7.46 (m, 3H), 7.24-7.31 (m, 4H), 2.28 (s, 3H). ^{13}C NMR (75.4 MHz, $CDCl_3$): δ 141.9, 139.6, 135.6, 133.4, 132.3, 130.4, 130.1, 128.0, 127.83, 127.77, 127.72, 127.5, 127.4, 126.2, 125.9, 20.4. MS (EI) m/z : 218 (M^+). IR (KBr, cm^{-1}): ν 3054, 1598, 1491, 1458, 1130, 1026, 857, 821, 757.

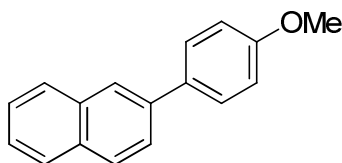


2-(m-tolyl)naphthalene.^[2] (3bc) The product was made following method B with a yield of 94% as a colorless oil. 1H NMR (300 MHz, $CDCl_3$): δ 8.02 (s, 1H),

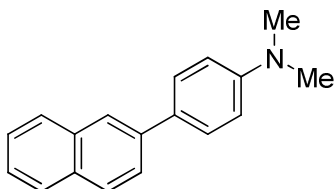
7.83-7.91(m, 3H), 7.73 (dd, $J = 1.8, 8.6$ Hz, 1H), 7.45-7.53 (m, 4H), 7.36 (t, $J = 7.5$ Hz, 1H), 7.19 (d, $J = 7.2$ Hz, 1H), 2.44 (s, 3H). ^{13}C NMR (75.4 MHz, CDCl_3): δ 141.2, 138.7, 138.5, 133.7, 132.7, 128.8, 128.4, 128.3, 128.24, 128.16, 127.7, 126.3, 125.9, 125.8, 125.7, 124.6, 21.5. MS (EI) m/z : 218 (M^+). IR (KBr, cm^{-1}): ν 3054, 1605, 1490, 856, 784, 745, 701.



2-(*p*-tolyl)naphthalene.^[3] (3bd) The product was made following method B with a yield of 93% as a white solid (An 87% yield was obtained in a 10 mmol scale in the presence of 1 mol% catalyst). ^1H NMR (300 MHz, CDCl_3): δ 7.98 (d, $J = 1.8$ Hz, 1H), 7.78-7.85 (m, 3H), 7.69 (dd, $J = 1.5, 8.4$ Hz, 1H), 7.58 (d, $J = 8.4$ Hz, 2H), 7.41-7.47 (m, 2H), 7.24 (d, $J = 7.8$ Hz, 2H), 2.37 (s, 3H). ^{13}C NMR (75.4 MHz, CDCl_3): δ 138.5, 138.2, 137.2, 133.8, 132.5, 129.6, 128.4, 128.2, 127.7, 127.3, 126.2, 125.8, 125.6, 125.4, 21.0. MS (EI) m/z : 218 (M^+). IR (KBr, cm^{-1}): ν 3058, 1502, 1265, 1131, 1019, 907, 810, 731.

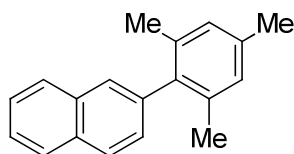


2-(4-methoxyphenyl)naphthalene.^[1] (3be) The product was made following method B with a yield of 88% as a white solid. ^1H NMR (300 MHz, CDCl_3): δ 7.97 (s, 1H), 7.80-7.87 (m, 3H), 7.69 (dd, $J = 1.8, 8.6$ Hz, 1H), 7.63 (d, $J = 8.4$ Hz, 2H), 7.41-7.49 (m, 2H), 6.99 (d, $J = 8.7$ Hz, 2H), 3.82 (s, 3H). ^{13}C NMR (75.4 MHz, CDCl_3): δ 159.3, 138.2, 133.8, 133.6, 132.4, 128.45, 128.39, 128.1, 127.7, 126.3, 125.7, 125.4, 125.0, 114.3, 55.3. MS (EI) m/z : 234 (M^+). IR (KBr, cm^{-1}): ν 2956, 1607, 1521, 1503, 1283, 1253, 1184, 1039, 814, 742

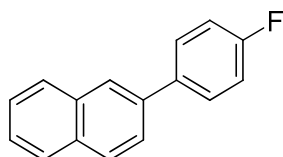


N,N-dimethyl-4-(naphthalen-2-yl)aniline.^[1] (3bf) The product was made following method B with a yield of 92% as a white solid. ^1H NMR (300 MHz, CDCl_3): δ 8.07 (s, 1H), 7.89-7.95 (m, 3H), 7.79-7.83 (m, 1H), 7.72 (d, $J = 8.7$ Hz, 2H), 7.47-7.57 (m, 2H), 6.91 (d, $J = 9.0$ Hz, 2H), 3.06 (s, 6H). ^{13}C NMR (75.4 MHz, CDCl_3): δ 150.1,

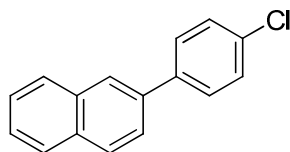
138.6, 133.9, 132.1, 129.0, 128.2, 128.0, 127.6, 126.1, 125.32, 125.26, 124.2, 112.9, 40.4. MS (EI) m/z : 247 (M^+). IR (KBr, cm^{-1}): ν 2850, 1607, 1528, 1503, 1354, 1231, 1065, 945, 857, 810, 742.



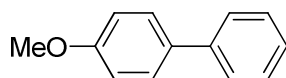
2-mesitylnaphthalene.^[4] (**3bg**) The product was made following method B with a yield of 91% as a colorless oil. ^1H NMR (300 MHz, CDCl_3): δ 7.80-7.90 (m, 3H), 7.60 (s, 1H), 7.47-7.50 (m, 2H), 7.28 (dd, J = 1.6, 8.3 Hz, 1H), 6.97 (s, 2H), 2.35 (s, 3H), 2.02 (s, 6H). ^{13}C NMR (75.4 MHz, CDCl_3): δ 138.9, 138.7, 136.8, 136.2, 133.6, 132.3, 128.2, 128.02, 127.96, 127.92, 127.8, 126.0, 125.7, 21.0, 20.7. MS (EI) m/z : 246 (M^+). IR (KBr, cm^{-1}): ν 2981, 1488, 1377, 1128, 1025, 895, 852, 820.



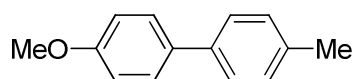
2-(4-fluorophenyl)naphthalene.^[1] (**3bh**) The product was made following method B with a yield of 72% as a white solid. ^1H NMR (300 MHz, CDCl_3): δ 7.92 (s, 1 H), 7.80-7.86 (m, 3 H), 7.58-7.64 (m, 3 H), 7.41-7.49 (m, 2 H), 7.09-7.14 (m, 2H). ^{13}C NMR (75.4 MHz, CDCl_3): δ 162.6 (d, $J_{\text{C-F}}$ = 244.6 Hz), 137.6, 137.3 (d, $J_{\text{C-F}}$ = 3.2 Hz), 133.7, 132.6, 129.0 (d, $J_{\text{C-F}}$ = 8.2 Hz), 128.6, 128.2, 127.7, 126.4, 126.0, 125.7, 125.4, 115.7 (d, $J_{\text{C-F}}$ = 20.6 Hz). MS (EI) m/z : 222 (M^+). IR (KBr, cm^{-1}): ν 3059, 1602, 1512, 1500, 1221, 1157, 1094, 822, 747.



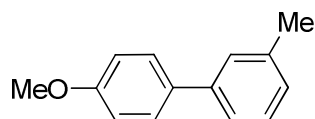
2-(4-chlorophenyl)naphthalene.^[5] (**3bi**) The product was made following method B with a yield of 43% as a white solid. ^1H NMR (300 MHz, CDCl_3): δ 7.96 (s, 1H), 7.82-7.88 (m, 3H), 7.64 (dd, J = 1.8, 8.4 Hz, 1H), 7.59 (d, J = 8.4 Hz, 2H), 7.46-7.51 (m, 2H), 7.41 (d, J = 8.4 Hz, 2H). ^{13}C NMR (75.4 MHz, CDCl_3): δ 139.6, 137.3, 133.7, 133.5, 132.7, 129.0, 128.7, 128.2, 127.7, 126.5, 126.2, 125.8, 125.2. MS (EI) m/z : 238 (M^+). IR (KBr, cm^{-1}): ν 2923, 1494, 1400, 1094, 1011, 816, 750.



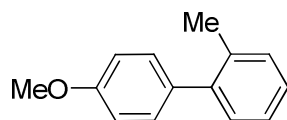
4-methoxybiphenyl.^[2] (**3fa**) The product was made following method B with a yield of 72% as a white solid. ¹H NMR (300 MHz, CDCl₃): δ 7.52-7.57 (m, 4H), 7.39-7.44 (m, 2H), 7.23-7.33 (m, 1H), 6.98 (d, *J* = 9.0 Hz, 2H), 3.84 (s, 3H). ¹³C NMR (75.4 MHz, CDCl₃): δ 159.0, 140.7, 133.7, 128.7, 128.1, 126.7, 126.6, 114.1, 55.2. MS (EI) *m/z*: 184 (M⁺). IR (KBr, cm⁻¹): ν 3003, 2961, 2837, 1608, 1519, 1487, 1246, 1183, 1043, 907, 834, 732.



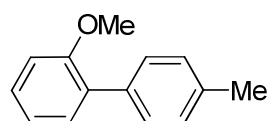
4-methoxy-4'-methylbiphenyl.^[6] (**3fb**) The product was made following method B with a yield of 75% as a white solid. ¹H NMR (300 MHz, CDCl₃): δ 7.50 (d, *J* = 8.7 Hz, 2H), 7.44 (d, *J* = 7.8 Hz, 2H), 7.21 (d, *J* = 7.8 Hz, 2H), 6.95 (d, *J* = 9.0 Hz, 2H), 3.81 (s, 3H), 2.36 (s, 3H). ¹³C NMR (75.4 MHz, CDCl₃): δ 158.9, 137.9, 136.3, 133.7, 129.4, 127.9, 126.5, 114.1, 55.2, 21.0. MS (EI) *m/z*: 198 (M⁺). IR (KBr, cm⁻¹): ν 2913, 1607, 1500, 1288, 1249, 1182, 1038, 907, 808, 732.



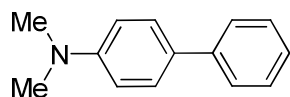
4'-methoxy-3-methylbiphenyl.^[7] (**3fc**) The product was made following method B with a yield of 73% as a white solid. ¹H NMR (300 MHz, CDCl₃): δ 7.52 (d, *J* = 8.4 Hz, 2H), 7.24-7.37 (m, 3H), 7.12 (d, *J* = 6.9 Hz, 1H), 6.96 (d, *J* = 8.4 Hz, 2H), 3.84 (s, 3H), 2.41 (s, 3H). ¹³C NMR (75.4 MHz, CDCl₃): δ 159.2, 140.9, 138.4, 134.0, 128.7, 128.2, 127.6, 127.5, 123.9, 114.2, 55.3, 21.4. MS (EI) *m/z*: 198 (M⁺). IR (KBr, cm⁻¹): ν 2911, 1603, 1515, 1484, 1248, 1186, 1030, 837, 785.



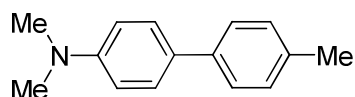
4'-methoxy-2-methylbiphenyl.^[7] (**3fd**) The product was made following method B with a yield of 75% as a colorless oil. ¹H NMR (300 MHz, CDCl₃): δ 7.21-7.26 (m, 6H), 6.95 (d, *J* = 9 Hz, 2H), 3.84 (s, 3H), 2.27 (s, 3H). ¹³C NMR (75.4 MHz, CDCl₃): δ 158.6, 141.6, 135.5, 134.4, 130.4, 130.3, 130.0, 127.0, 125.8, 113.5, 55.2, 20.4. MS (EI) *m/z*: 198 (M⁺). IR (KBr, cm⁻¹): ν 2954, 2931, 2835, 1621, 1516, 1483, 1464, 1243, 1176, 1038, 833, 760, 731.



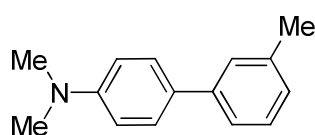
2-methoxy-4'-methylbiphenyl.^[8] (**3gb**) The product was made following method B with a yield of 75% as a white solid. ¹H NMR (300 MHz, CDCl₃): δ 7.43 (d, *J* = 7.8 Hz, 2H), 7.20-7.32 (m, 4H), 6.94-7.03 (m, 2H), 3.77 (s, 3H), 2.37 (s, 3H). ¹³C NMR (75.4 MHz, CDCl₃): δ 156.5, 136.5, 135.6, 130.73, 130.65, 129.4, 128.7, 128.3, 120.7, 111.1, 55.4, 21.2. MS (EI) *m/z*: 198 (M⁺). IR (KBr, cm⁻¹) : ν 2920, 1597, 1486, 1260, 1235, 1180, 1123, 1029, 819, 753.



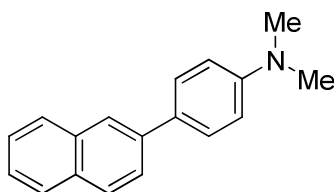
N,N-dimethylbiphenyl-4-amine.^[9] (**3ha**) The product was made following method B with a yield of 83% as a yellow oil. ¹H NMR (300 MHz, CDCl₃): δ 7.49-7.57 (m, 4H), 7.36-7.41 (m, 2H), 7.23-7.27 (m, 1H), 6.81 (d, *J* = 9 Hz, 3H), 3.06 (s, 3H). ¹³C NMR (75.4 MHz, CDCl₃): δ 150.0, 141.3, 129.4, 128.7, 127.8, 126.4, 126.0, 112.9, 40.5. ESI-MS *m/z*: 198.1 [M+H]⁺. IR (KBr, cm⁻¹) : ν 2917, 1719, 1610, 1507, 1369, 1286, 911, 812.



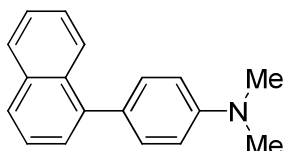
N,N,4'-trimethylbiphenyl-4-amine.^[10] (**3hb**) The product was made following method B with a yield of 81% as a white solid. ¹H NMR (300 MHz, CDCl₃): δ 7.43-7.49 (m, 4H), 7.19 (d, *J* = 8.1 Hz, 2H), 6.78 (d, *J* = 8.4 Hz, 2H), 2.95 (s, 6H), 2.36 (s, 3H). ¹³C NMR (75.4 MHz, CDCl₃): δ 149.9, 138.4, 135.6, 129.41, 129.35, 127.5, 126.2, 112.8, 40.5, 20.9. ESI-MS *m/z*: 212.1 [M+H]⁺. IR (KBr, cm⁻¹) : ν 2916, 1681, 1607, 1503, 1351, 1220, 1160, 1114, 946, 807, 731.



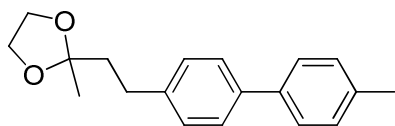
N,N,3'-trimethylbiphenyl-4-amine. (**3hc**) The product was made following method B with a yield of 88% as a yellow oil. ¹H NMR (300 MHz, CDCl₃): δ 7.50 (d, *J* = 9.0 Hz, 2H), 7.28-7.37 (m, 3H), 7.08 (d, *J* = 7.8 Hz, 1H), 6.81 (d, *J* = 9.0 Hz, 2H), 2.97 (s, 6H), 2.39 (s, 3H). ¹³C NMR (75.4 MHz, CDCl₃): δ 150.0, 141.3, 138.2, 129.5, 128.6, 127.8, 127.2, 126.8, 123.5, 112.8, 40.5, 21.5. ESI-MS *m/z*: 212.1 [M+H]⁺. IR (KBr, cm⁻¹) : ν 2918, 1678, 1605, 1525, 1485, 1444, 1352, 1225, 1200, 1166, 946, 819, 781.



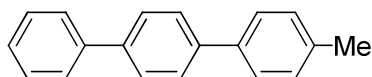
N,N-dimethyl-4-(naphthalen-2-yl)aniline.^[1] (3hj) The product was made following method B with a yield of 73% as a white solid. ¹H NMR (300 MHz, CDCl₃): δ 8.13 (s, 1H), 7.95-8.00 (m, 3H), 7.85-7.89 (m, 1H), 7.78 (d, *J* = 8.7 Hz, 2H), 7.53-7.63 (m, 2H), 6.95 (d, *J* = 9.0 Hz, 2H), 3.01 (s, 6H). ¹³C NMR (75.4 MHz, CDCl₃): δ 150.0, 138.6, 133.9, 132.1, 128.9, 128.2, 128.0, 127.6, 126.1, 125.29, 125.24, 124.1, 112.8, 40.3. MS (EI) *m/z*: 247 (M⁺). IR (KBr, cm⁻¹): ν 2850, 1607, 1528, 1503, 1354, 1231, 1065, 945, 857, 810, 742.



N,N-dimethyl-4-(naphthalen-1-yl)aniline.^[11] (3hk) The product was made following method B with a yield of 61% as a colorless oil. ¹H NMR (300 MHz, CDCl₃): δ 8.02 (d, *J* = 7.5 Hz, 1H), 7.85 (d, *J* = 8.1 Hz, 1H), 7.77 (d, *J* = 8.1 Hz, 1H), 7.37-7.49 (m, 6H), 6.82 (d, *J* = 8.7 Hz, 2H), 2.97 (s, 3H). ¹³C NMR (75.4 MHz, CDCl₃): δ 149.8, 140.5, 134.0, 132.0, 130.8, 128.8, 128.2, 126.9, 126.8, 126.4, 125.7, 125.6, 125.5, 112.3, 40.5. MS (EI) *m/z*: 247 (M⁺). IR (KBr, cm⁻¹): ν 3040, 2848, 1395, 1352, 1197, 800, 776.

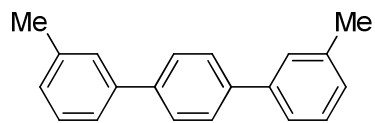


2-methyl-2-(2-(4'-methylbiphenyl-4-yl)ethyl)-1,3-dioxolane. (16) The product was made following method B with a yield of 66% as a white solid. ¹H NMR (300 MHz, CDCl₃): δ 7.47-7.52 (m, 4H), 7.22-7.28 (m, 4H), 3.97-4.01 (m, 4H), 2.73-2.78 (m, 2H), 2.38 (s, 3H), 1.97-2.03 (m, 2H), 1.39 (s, 3H). ¹³C NMR (75.4 MHz, CDCl₃): δ 141.1, 138.7, 138.3, 136.8, 129.5, 128.7, 127.0, 126.9, 109.7, 64.7, 40.9, 29.7, 23.9, 20.9. ESI-MS *m/z*: 305.1506 [M+Na]⁺, calcd. for C₁₉H₂₂NaO₂: 305.1517. IR (KBr, cm⁻¹): ν 2978, 2929, 2874, 1498, 1373, 1252, 1221, 1140, 1057, 862, 808.



4-(4-Methylphenyl)biphenyl.^[12] (19) The product was made following method B with a yield of 75% as a white solid. ¹H NMR (300 MHz, CDCl₃): δ 7.66 (s, 4H), 7.64 (d, *J* = 7.2 Hz, 2H), 7.54 (d, *J* = 8.1 Hz, 2H), 7.45 (t, *J* = 7.5 Hz, 2H), 7.35 (t, *J* = 7.5 Hz, 1H), 7.26 (d, *J* = 8.1 Hz, 2H), 2.40 (s, 3H). ¹³C NMR (75.4 MHz, CDCl₃): δ 140.8, 140.1, 139.9, 137.8, 137.2, 129.6, 128.9, 127.5, 127.3, 127.1, 126.9, 21.0. MS (EI) *m/z*: 244 (M⁺). IR (KBr, cm⁻¹): ν 2918, 2849, 1633, 1485, 813, 760.

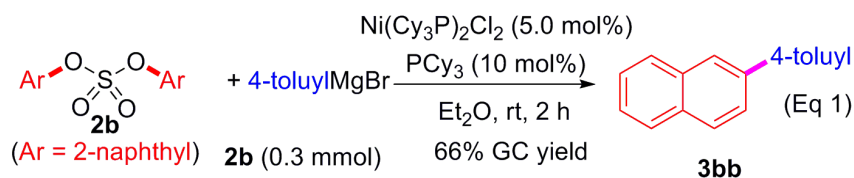
Byproduct of 3fc



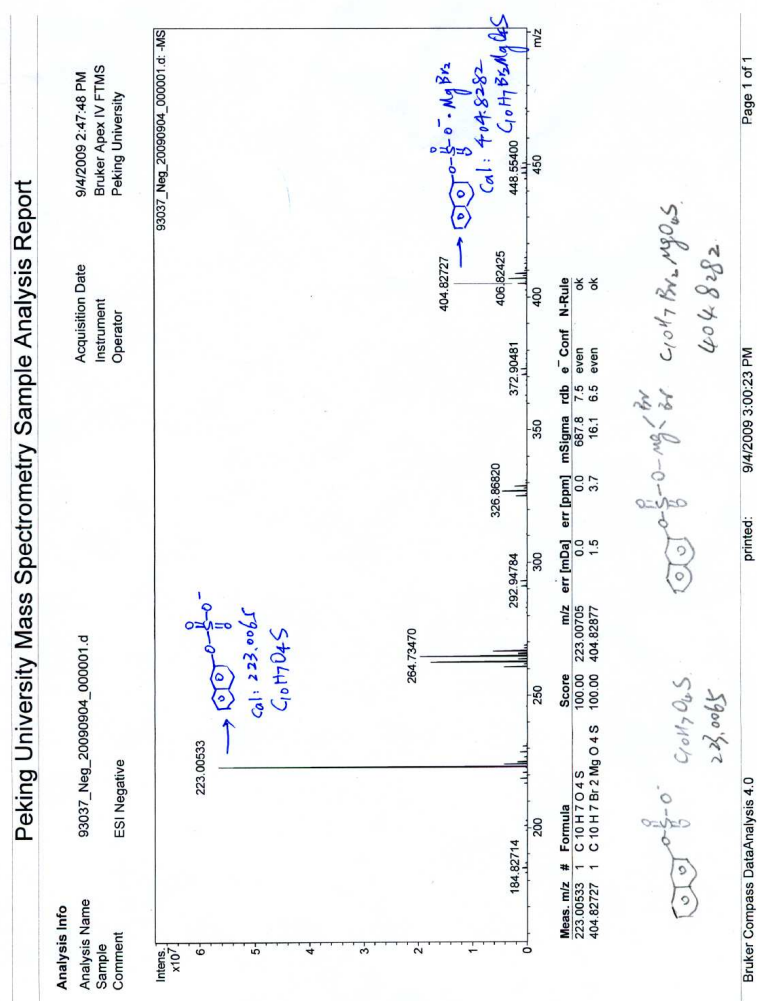
3,3''-Dimethyl-p-terphenyl. The product was made following method B with a yield of 17% as a white solid. ^1H NMR (300 MHz, CDCl_3): δ 7.65 (s, 4H), 7.4-.45 (m, 4H), 7.34 (t, $J = 7.5$ Hz, 2H), 7.17(d, $J = 7.8$ Hz, 2H), 2.42 (s, 6H). ^{13}C NMR (75.4 MHz, CDCl_3): δ 140.8, 140.2, 138.5, 128.8, 128.1, 127.9, 127.5, 124.2, 21.5. MS (EI) m/z : 258 (M^+). IR (KBr, cm^{-1}): ν 3053, 1607, 1482, 1265, 909, 839, 781, 737.

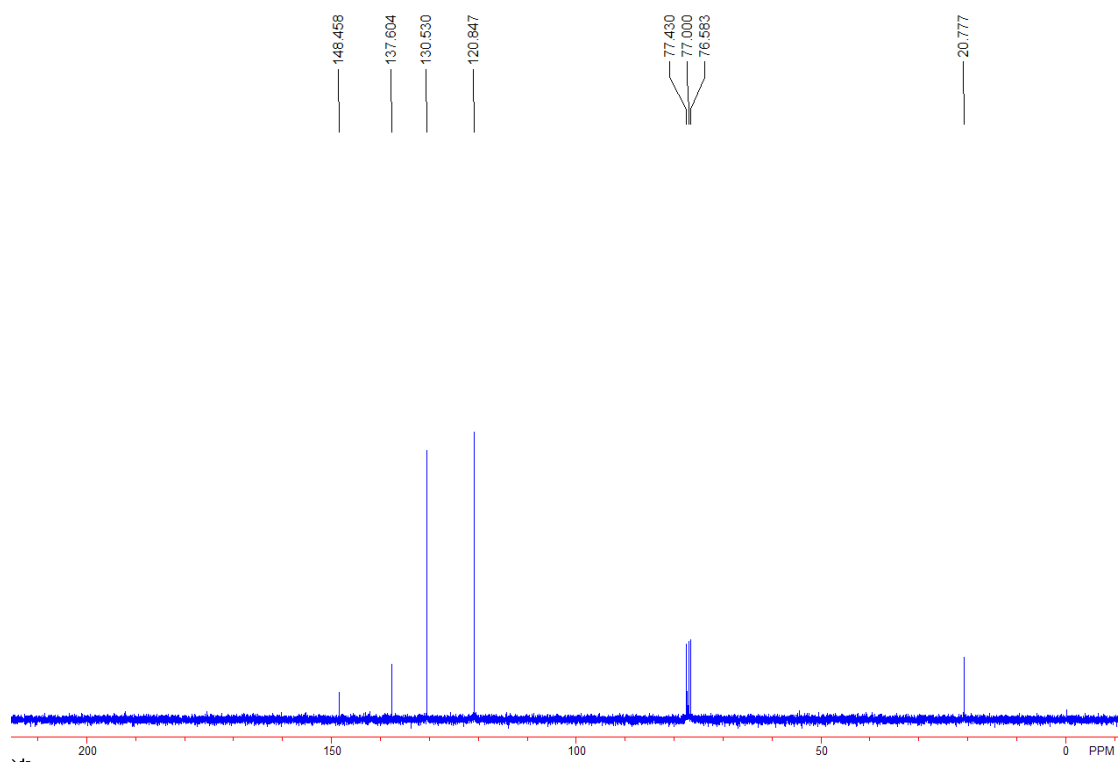
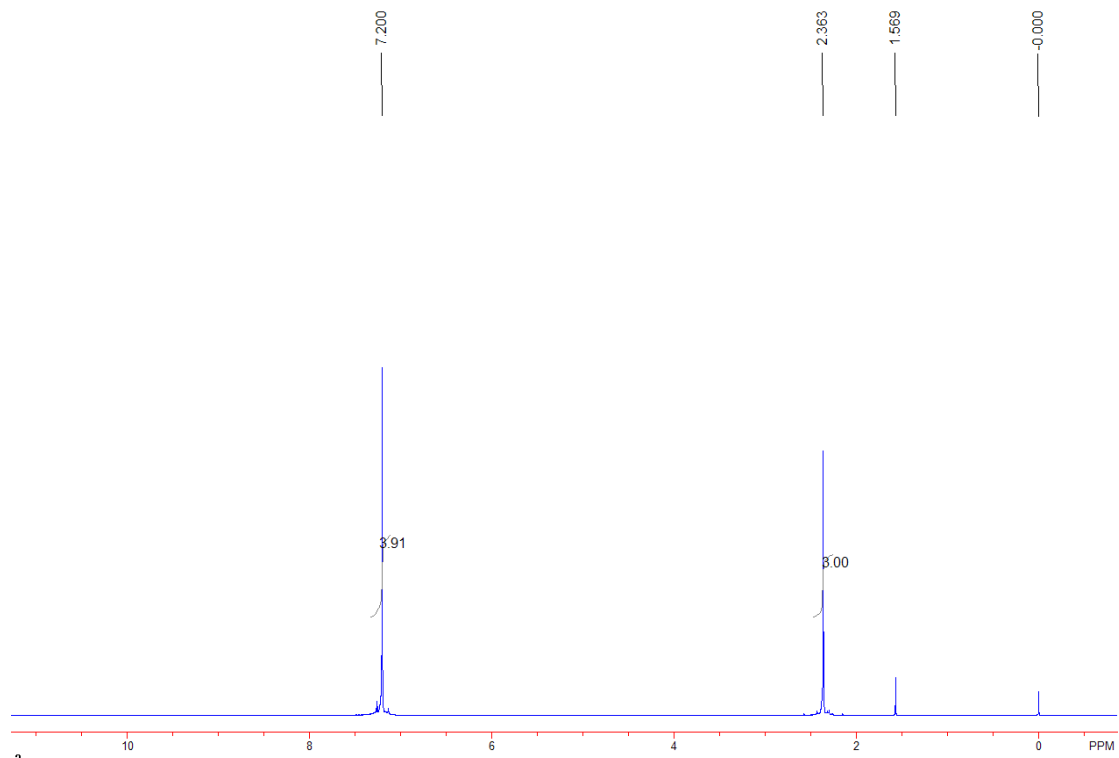
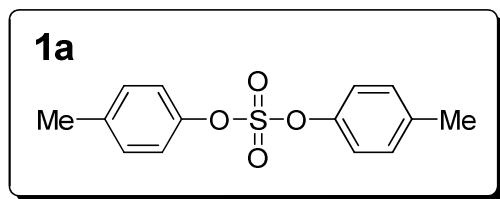
Mechanism exploration

Eq S1

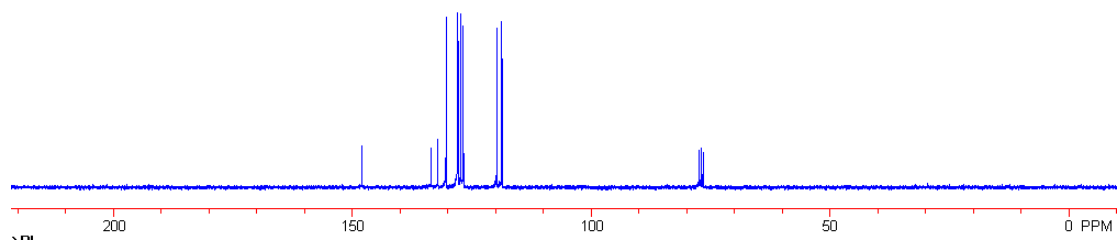
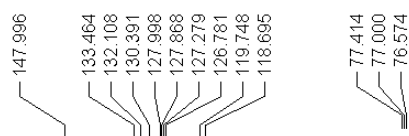
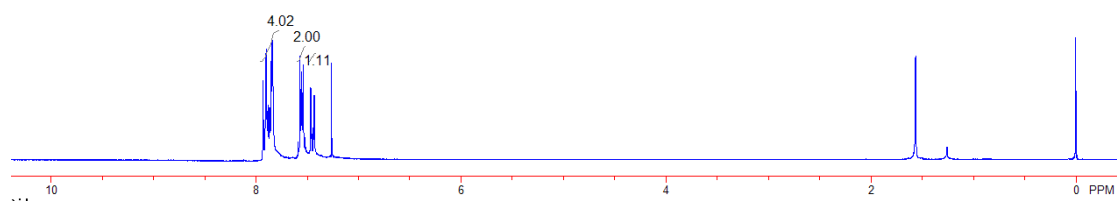
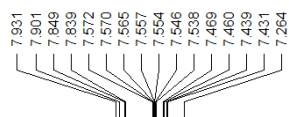
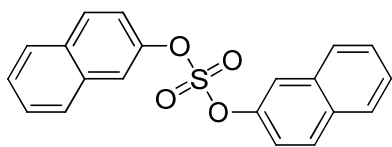


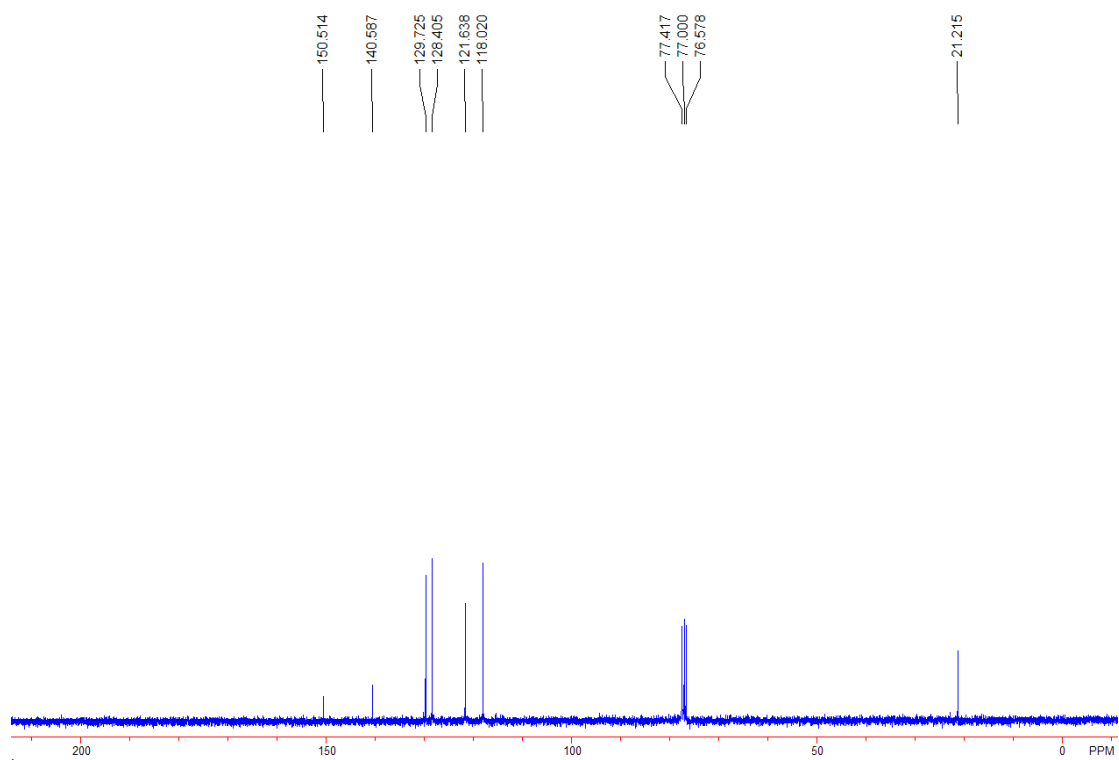
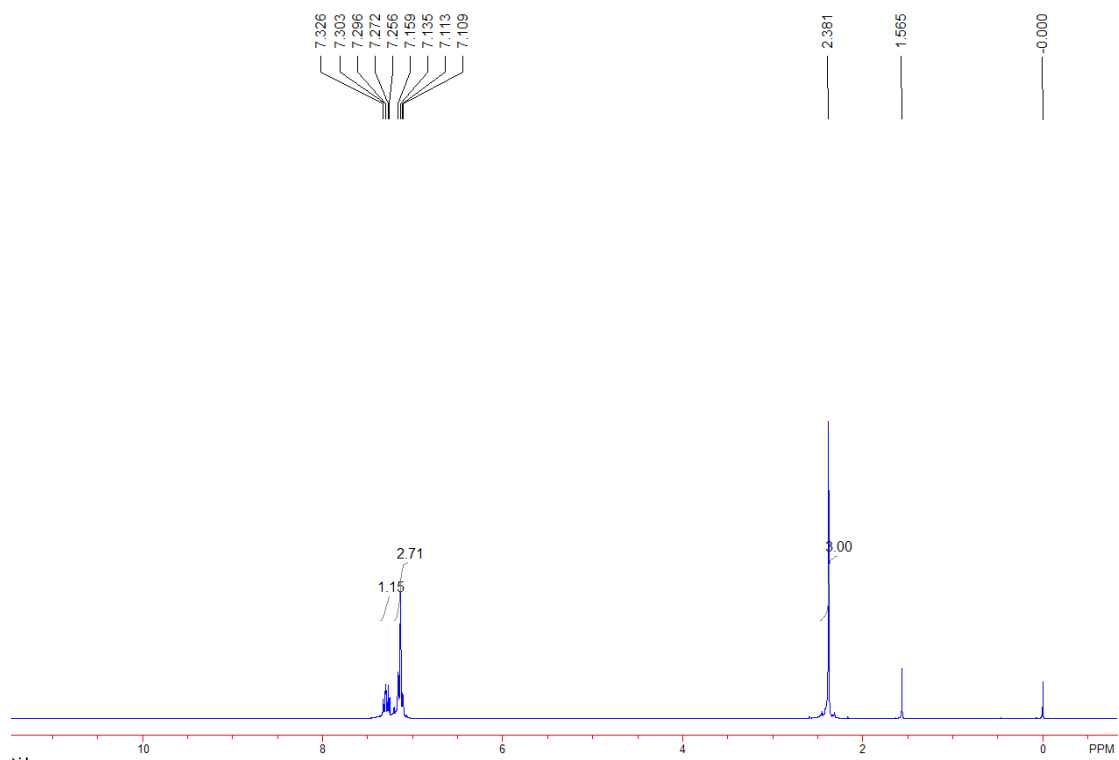
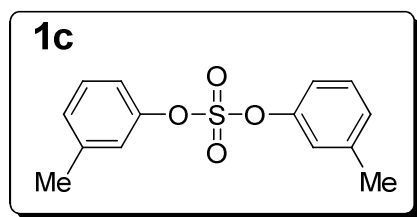
ESI MS spectroscopy of the monosulfate species

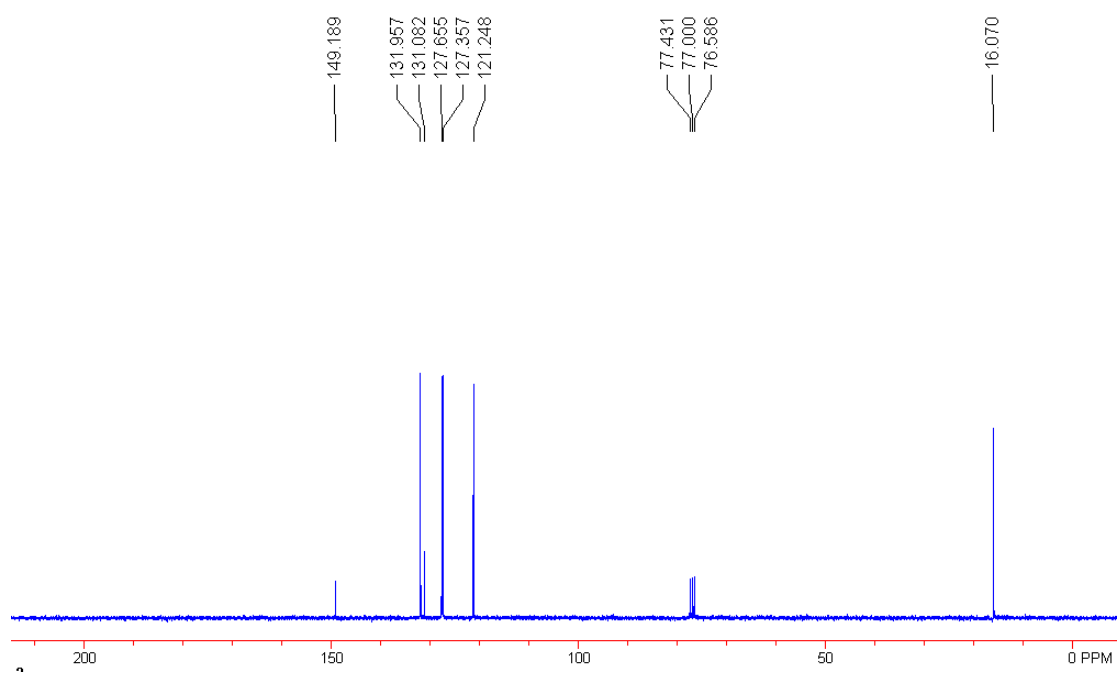
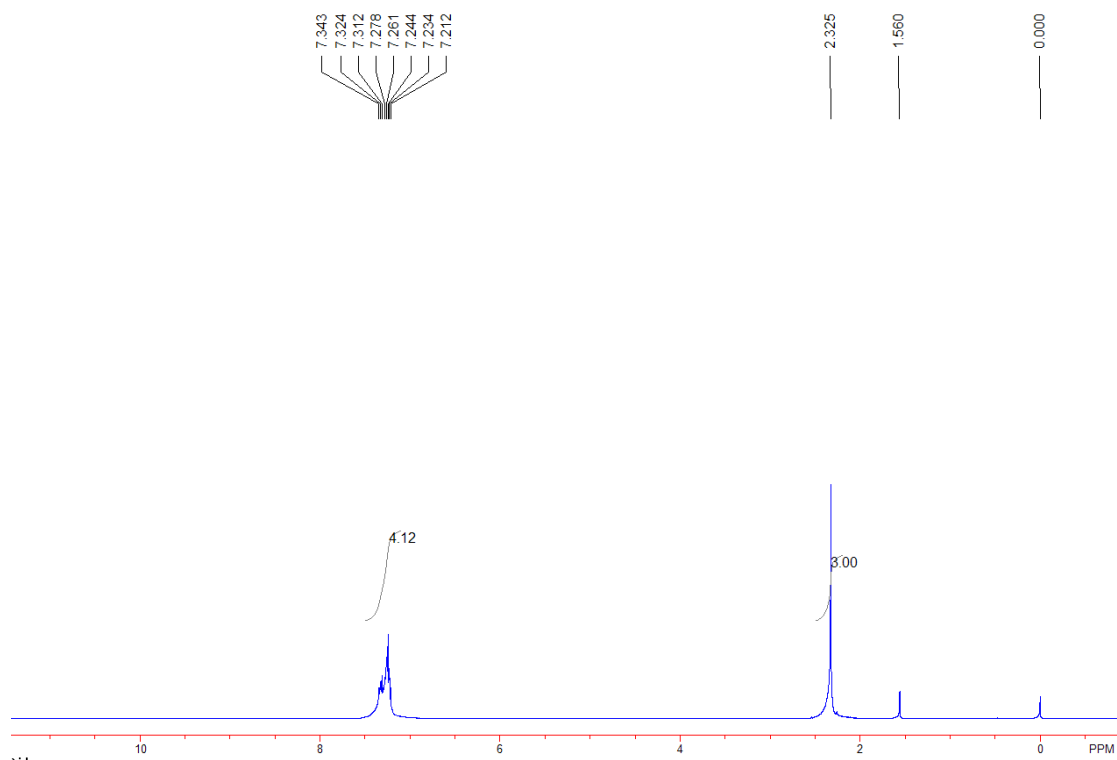
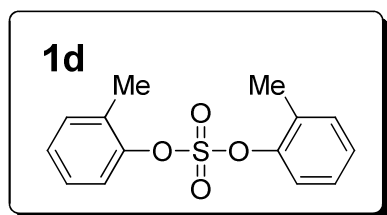


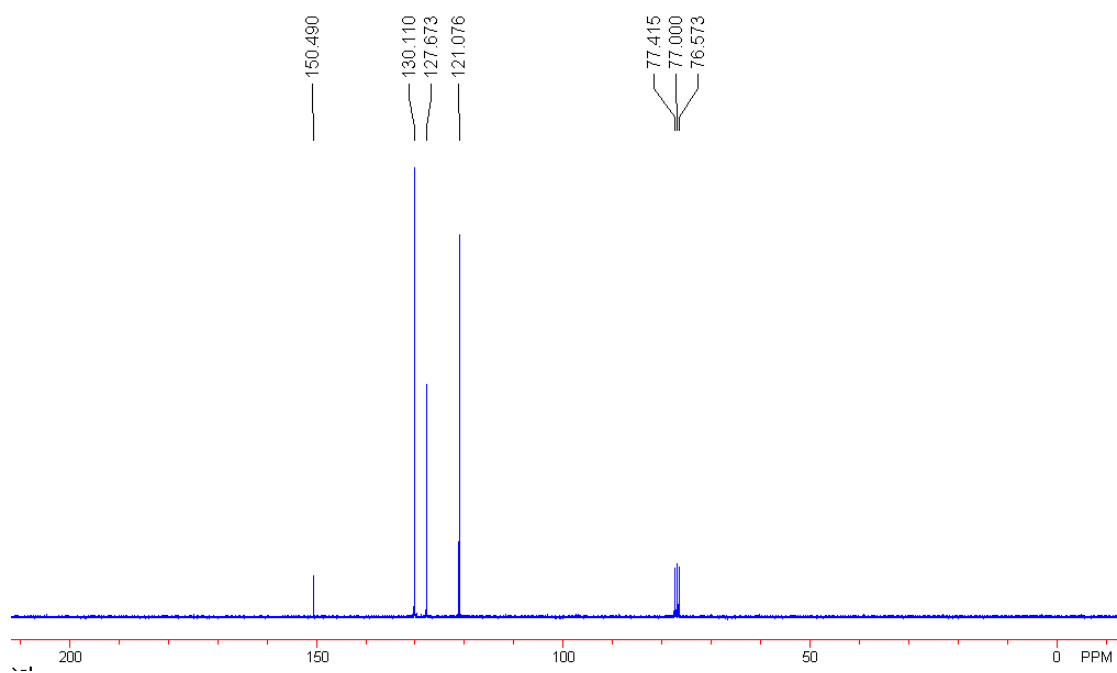
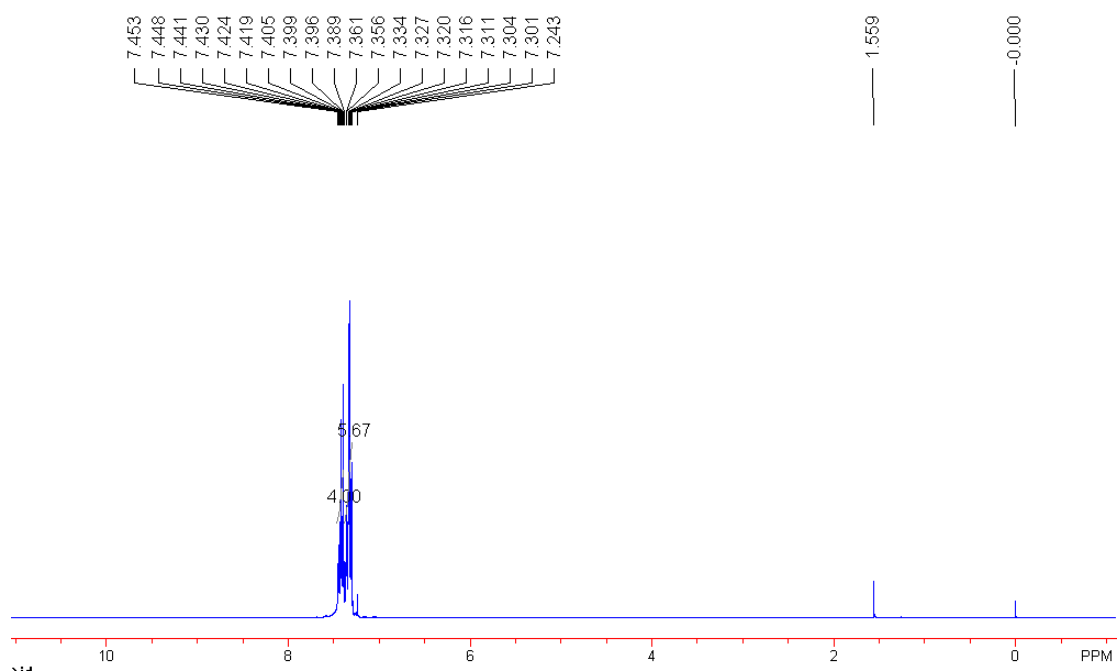
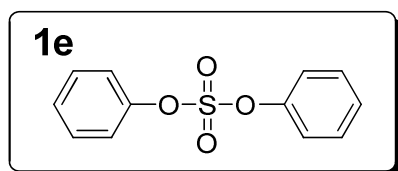


1b

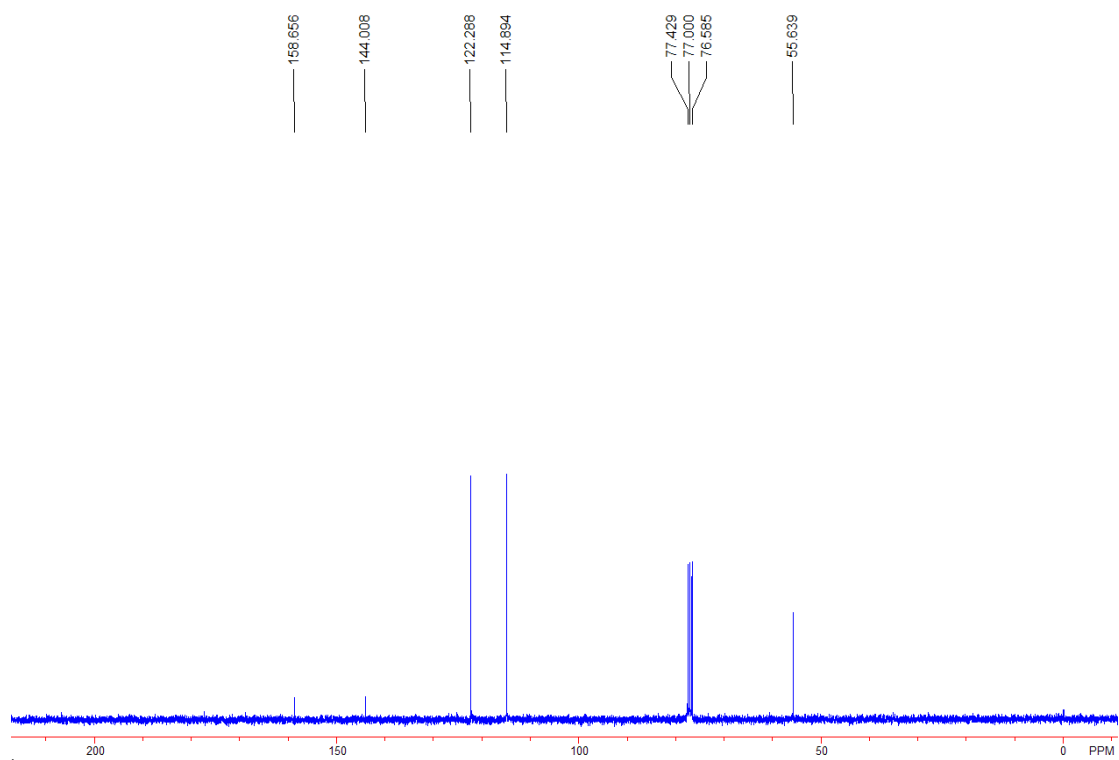
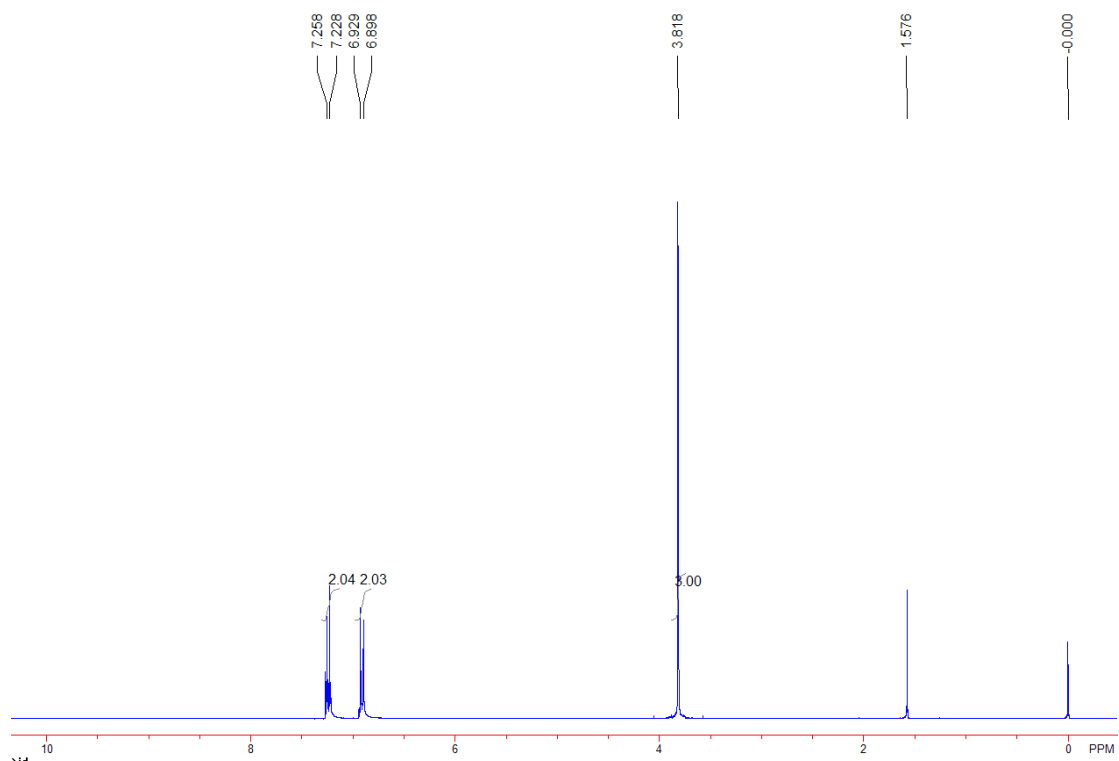
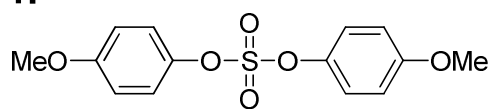


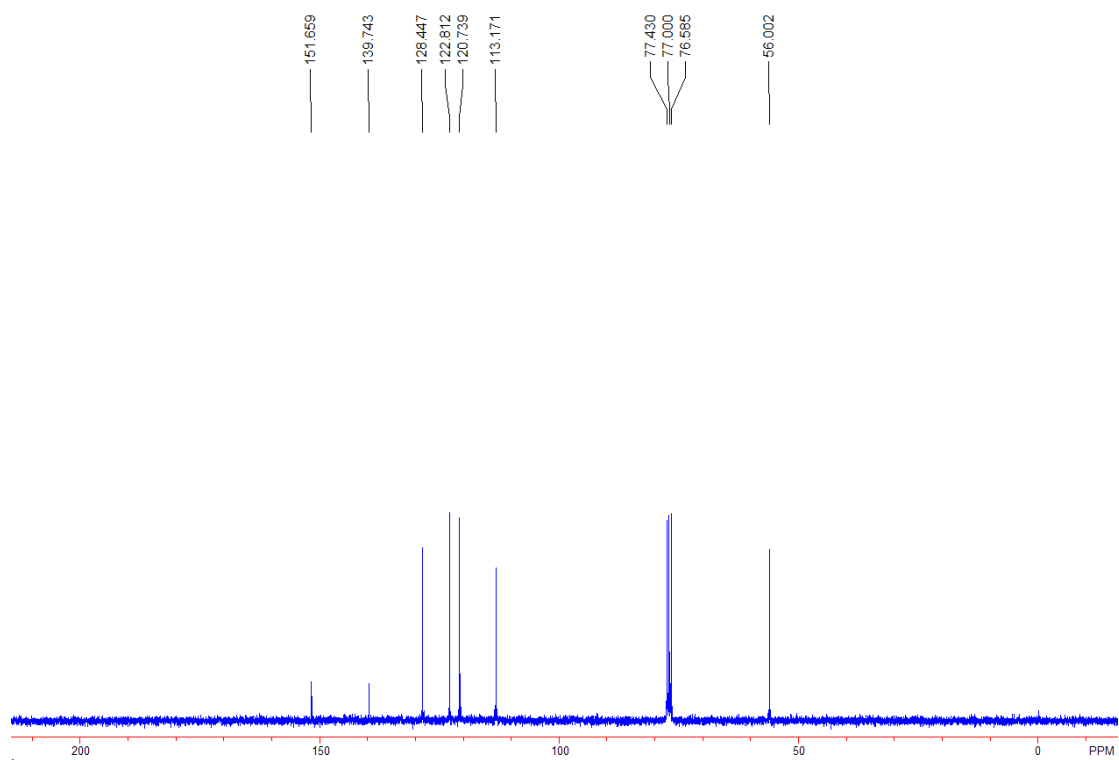
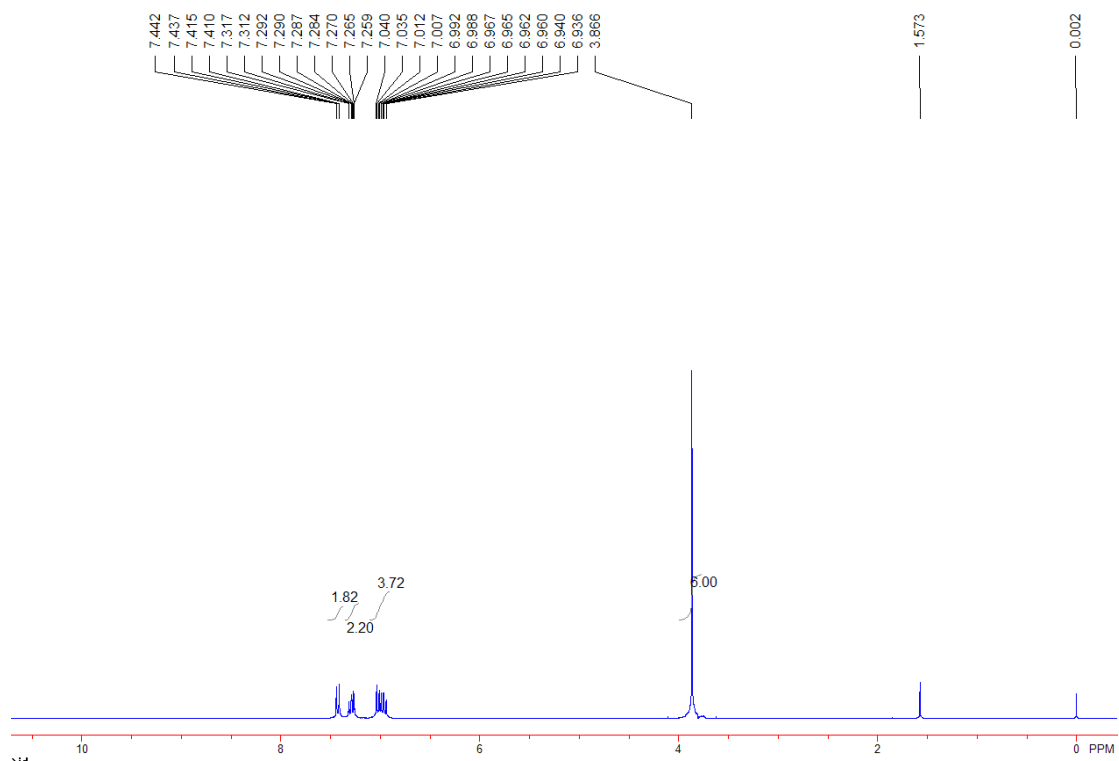
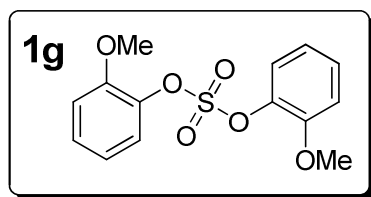




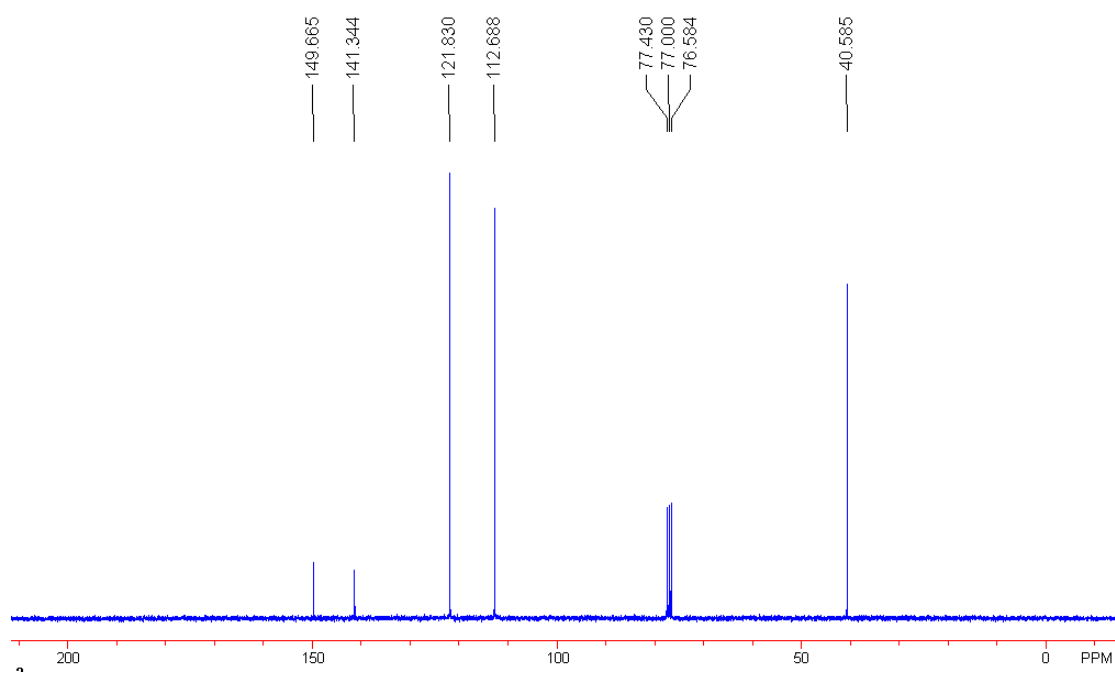
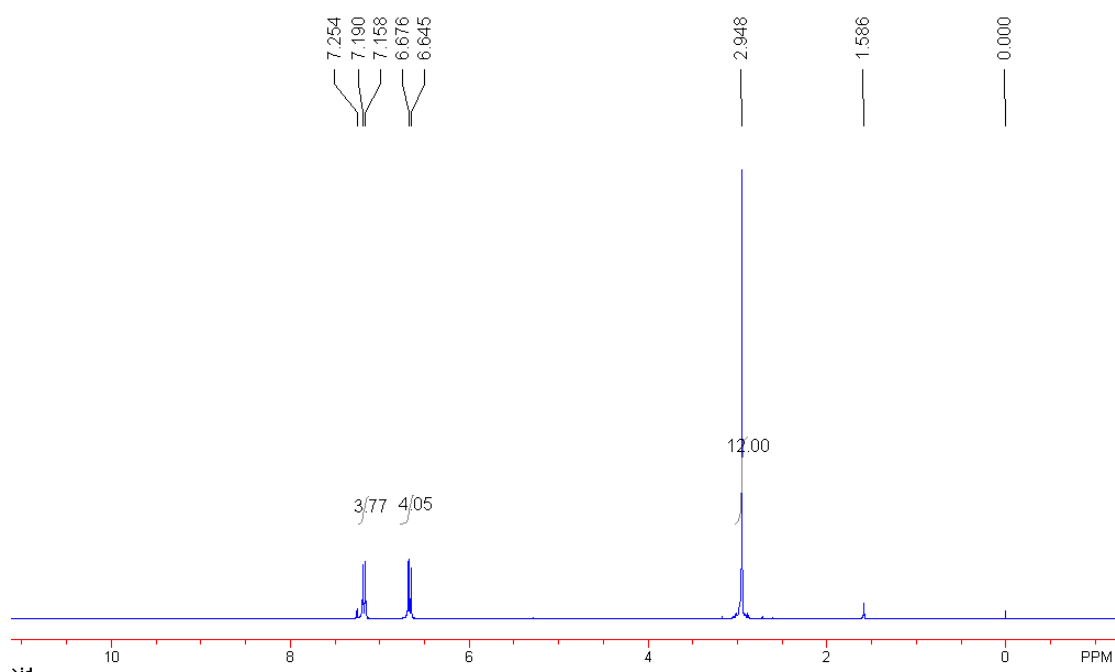
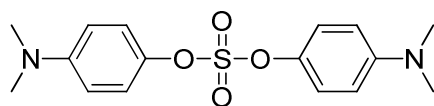


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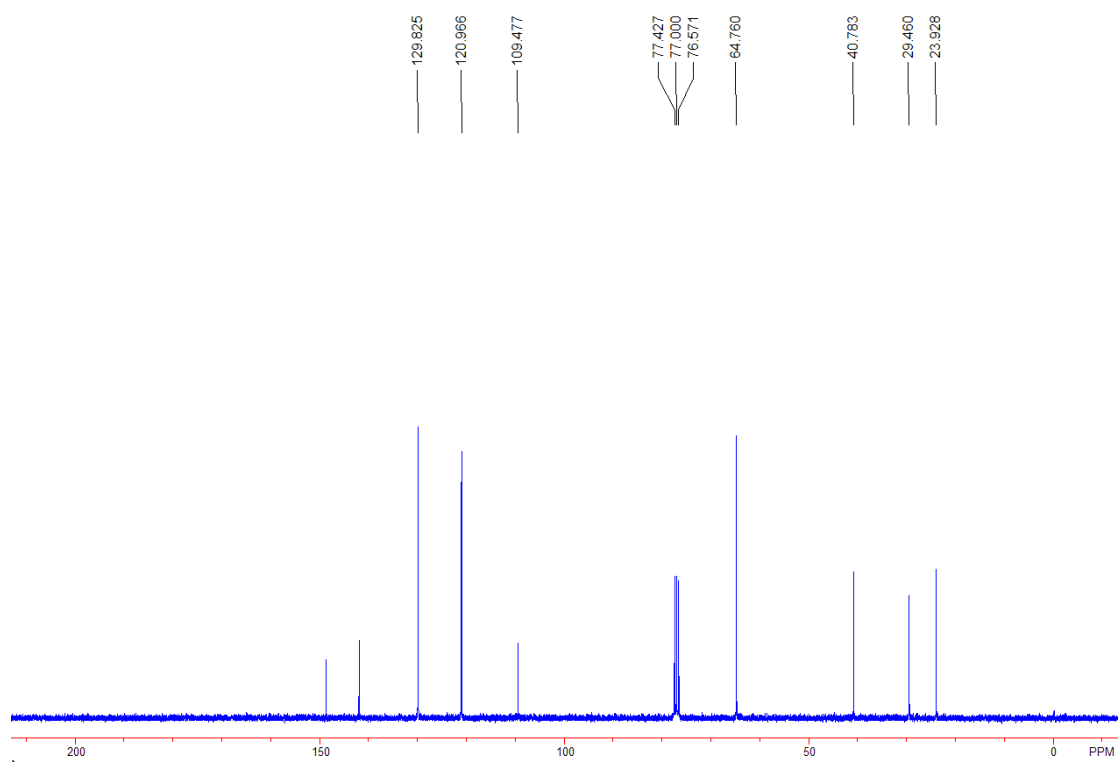
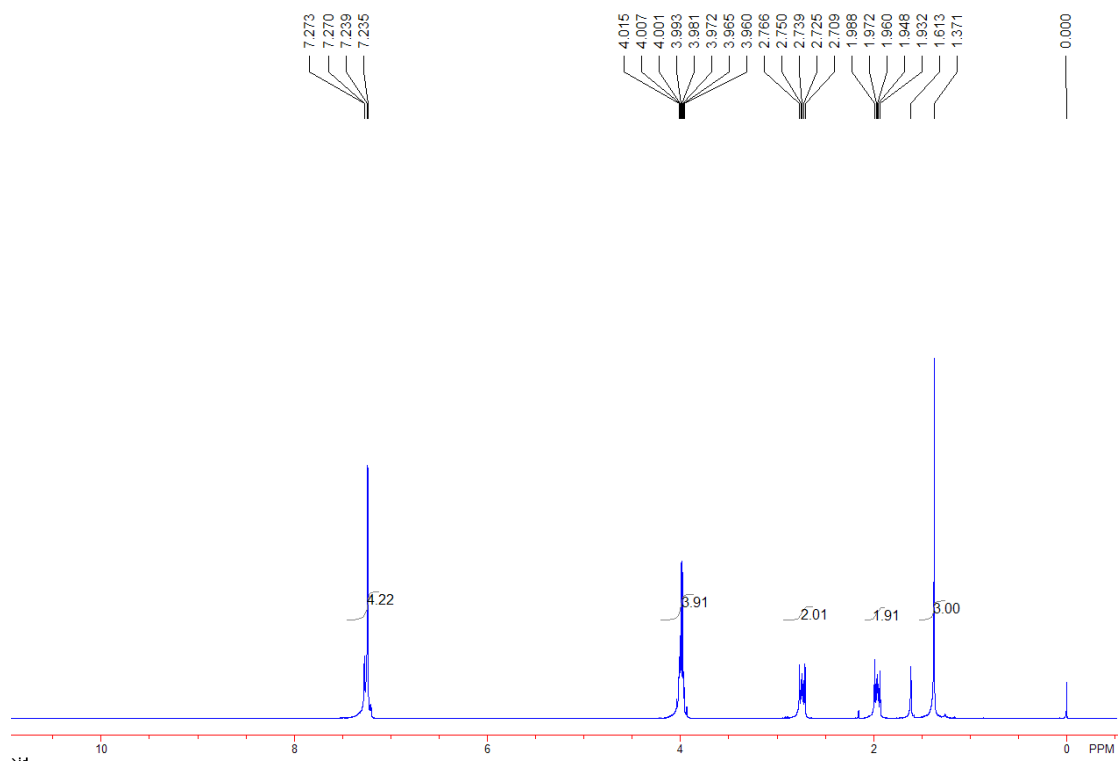
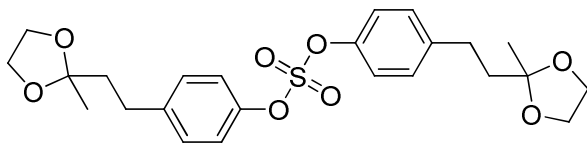




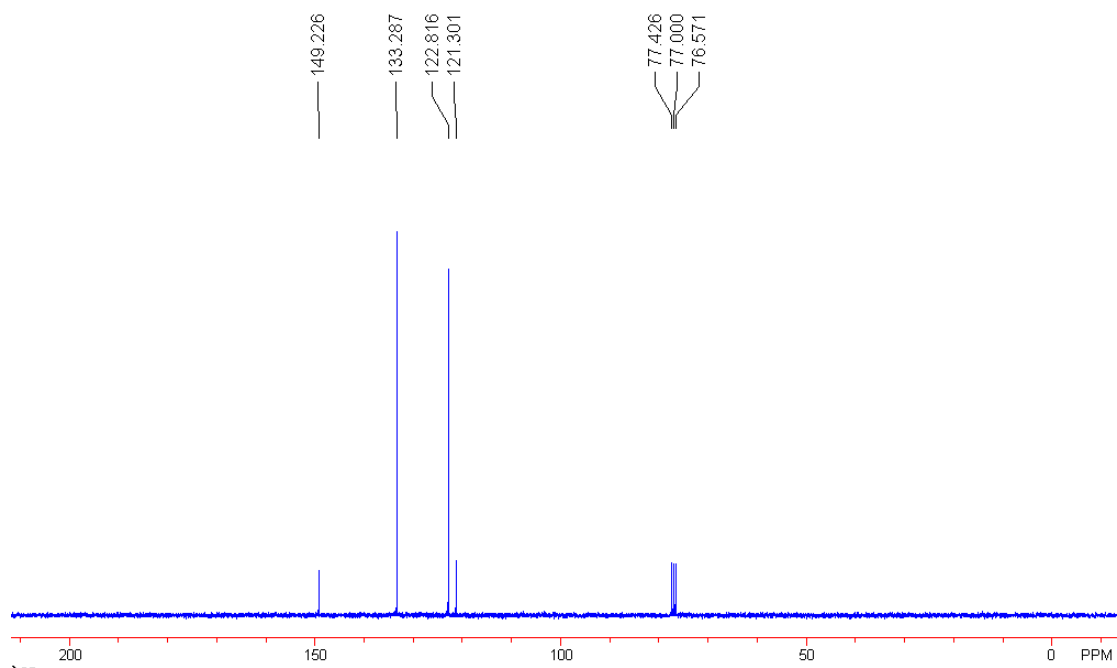
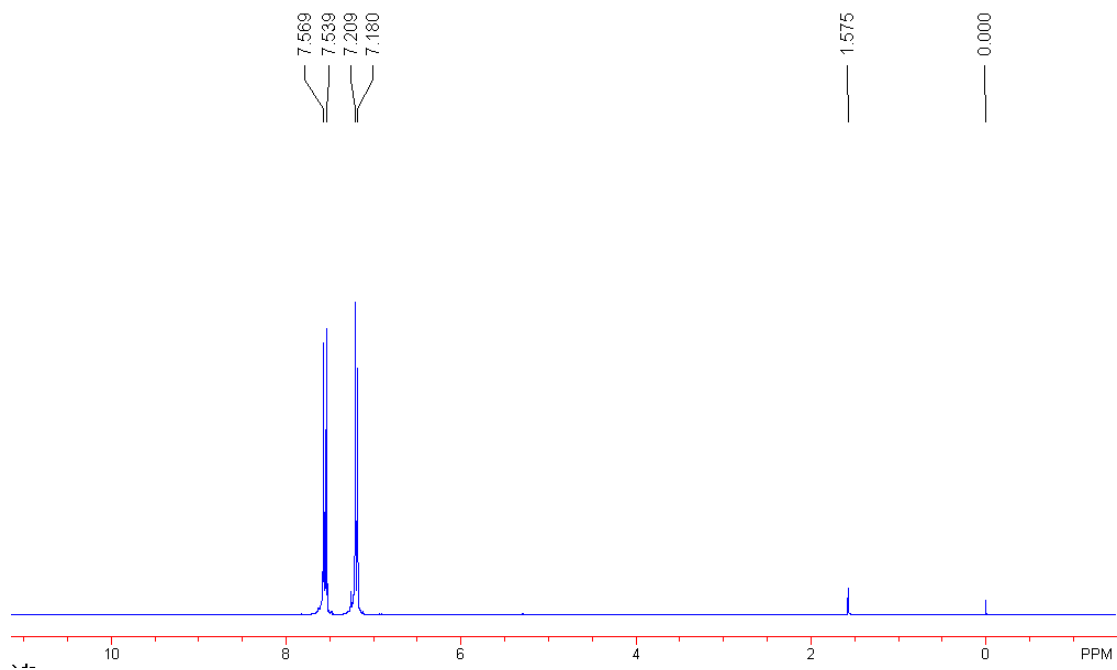
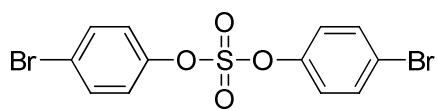
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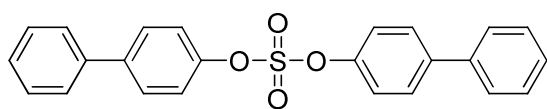
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17



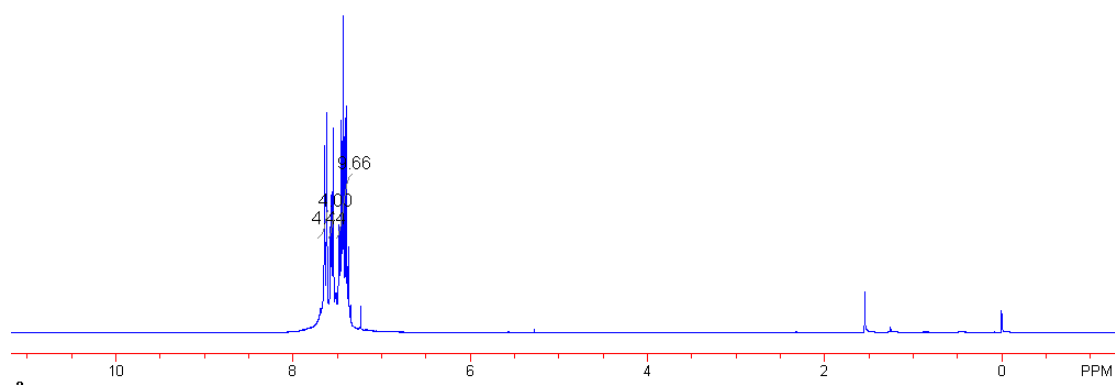
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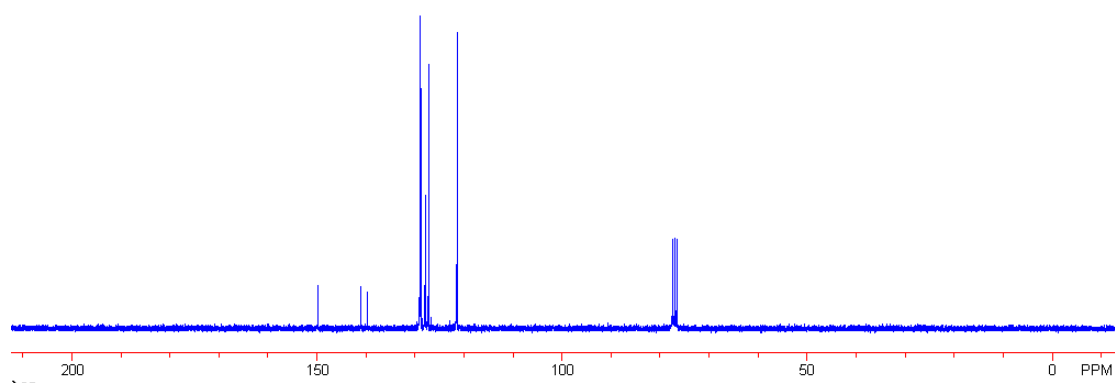
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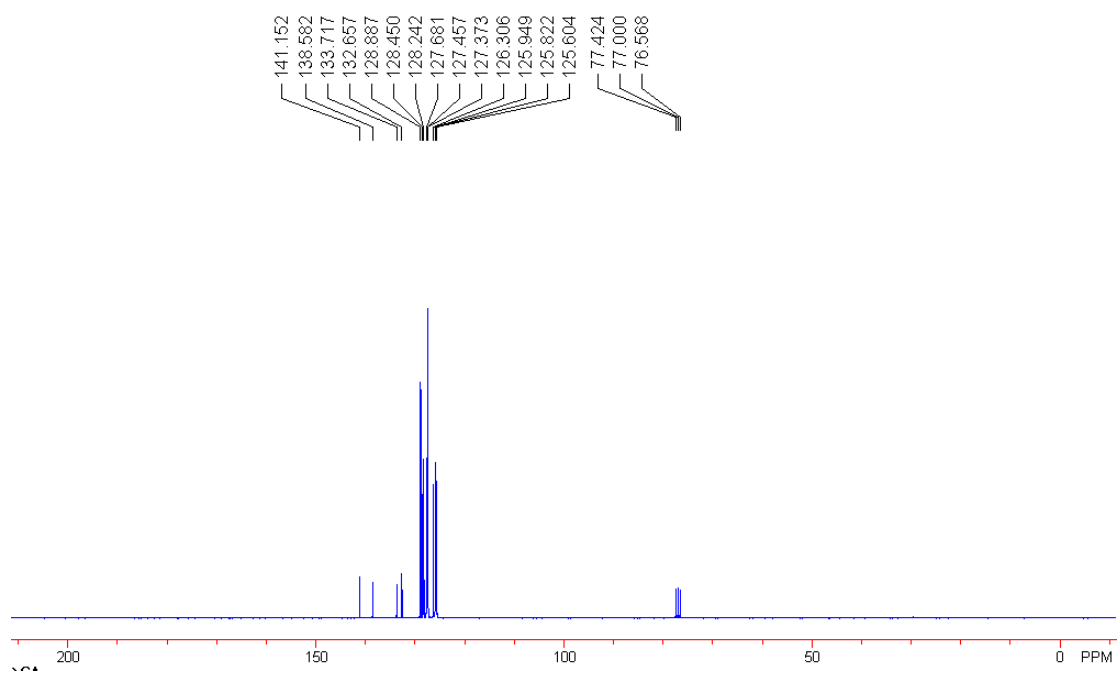
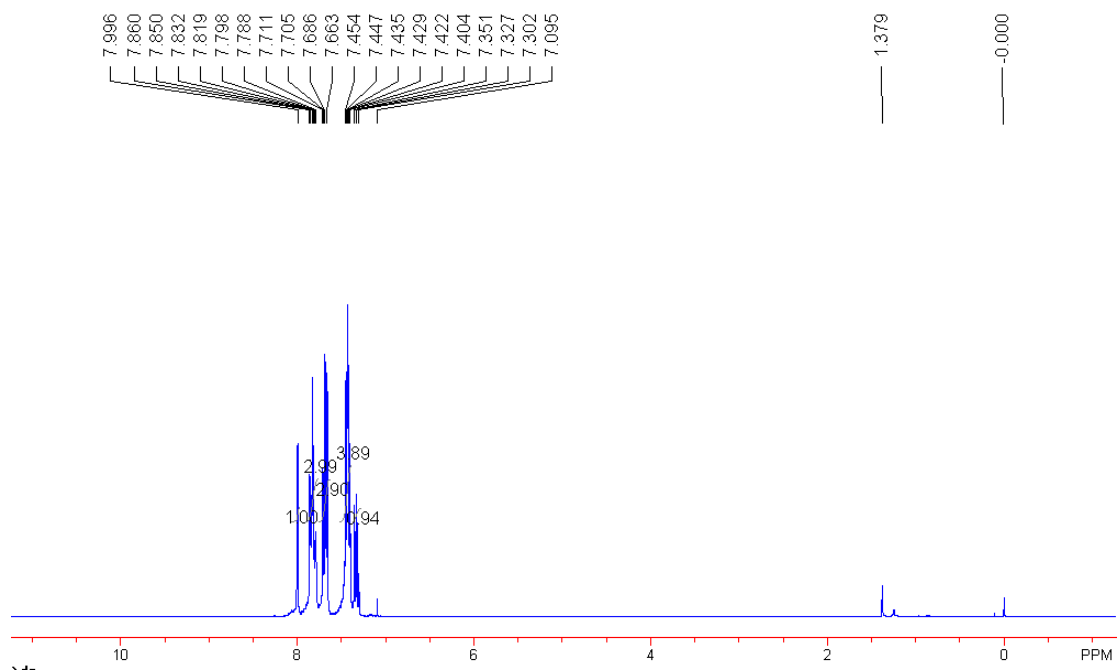
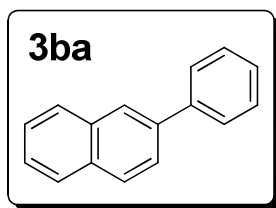
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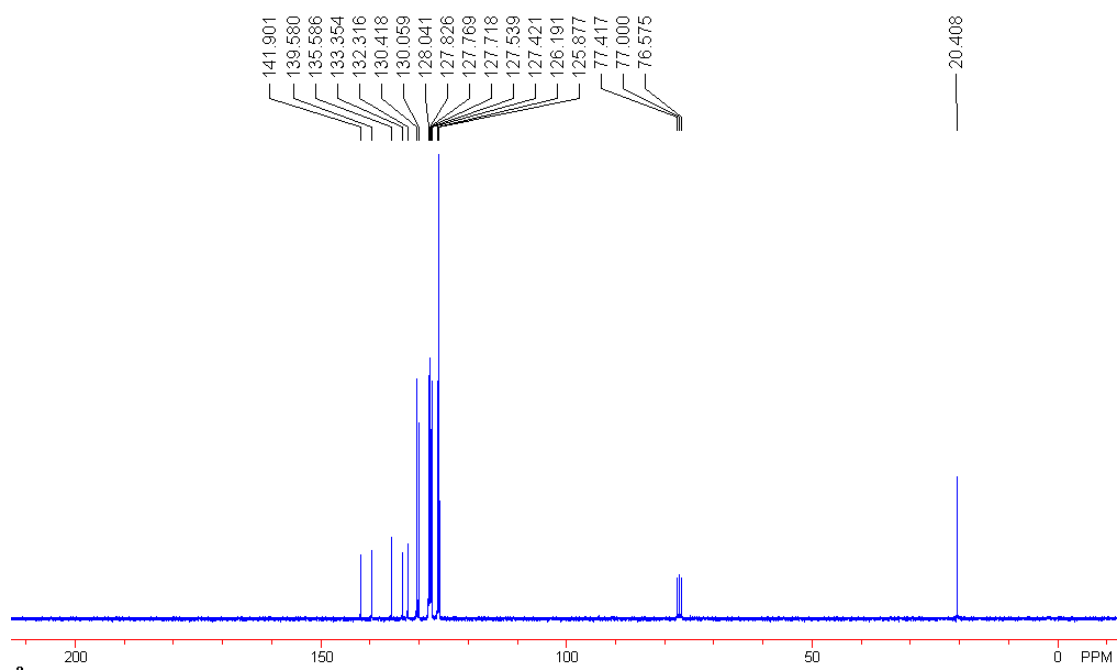
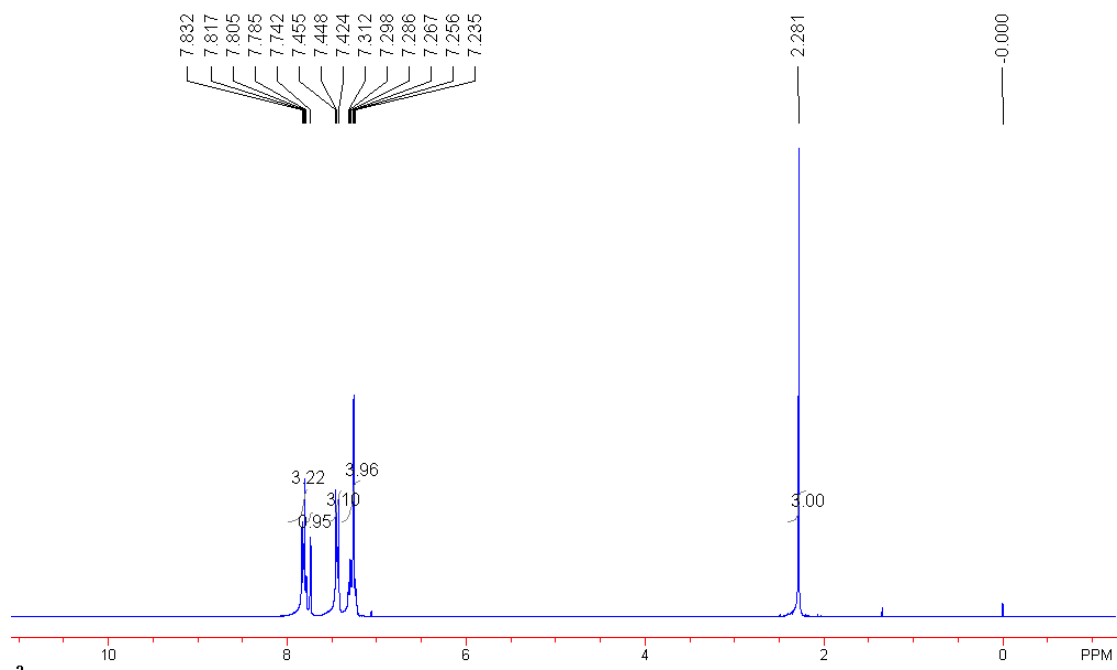
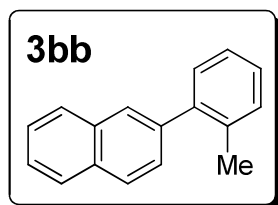


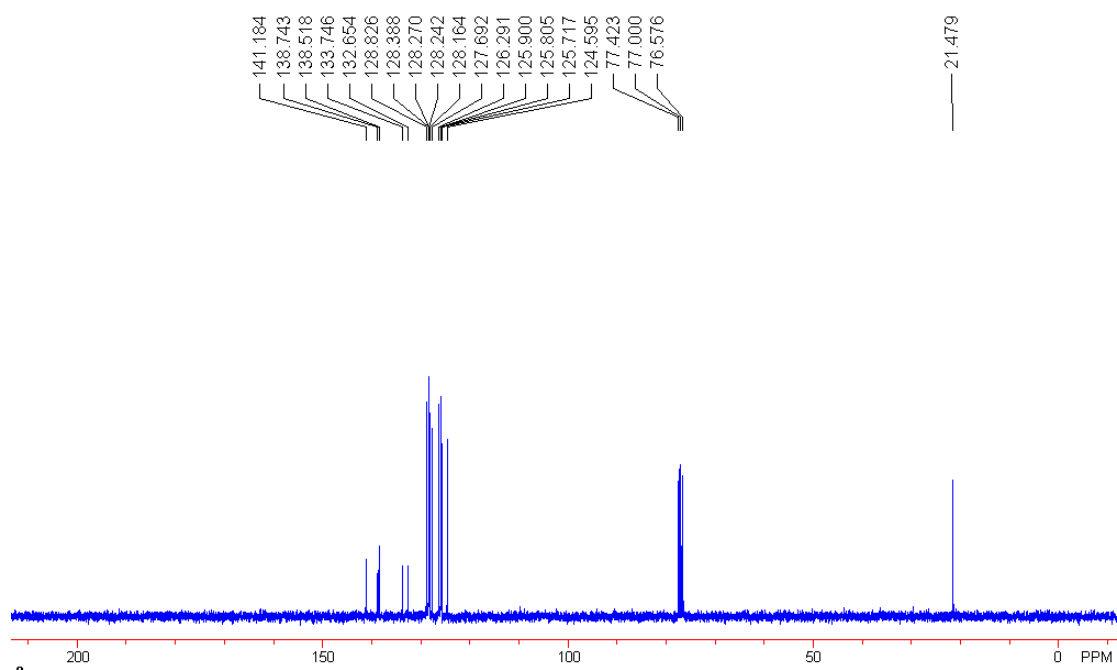
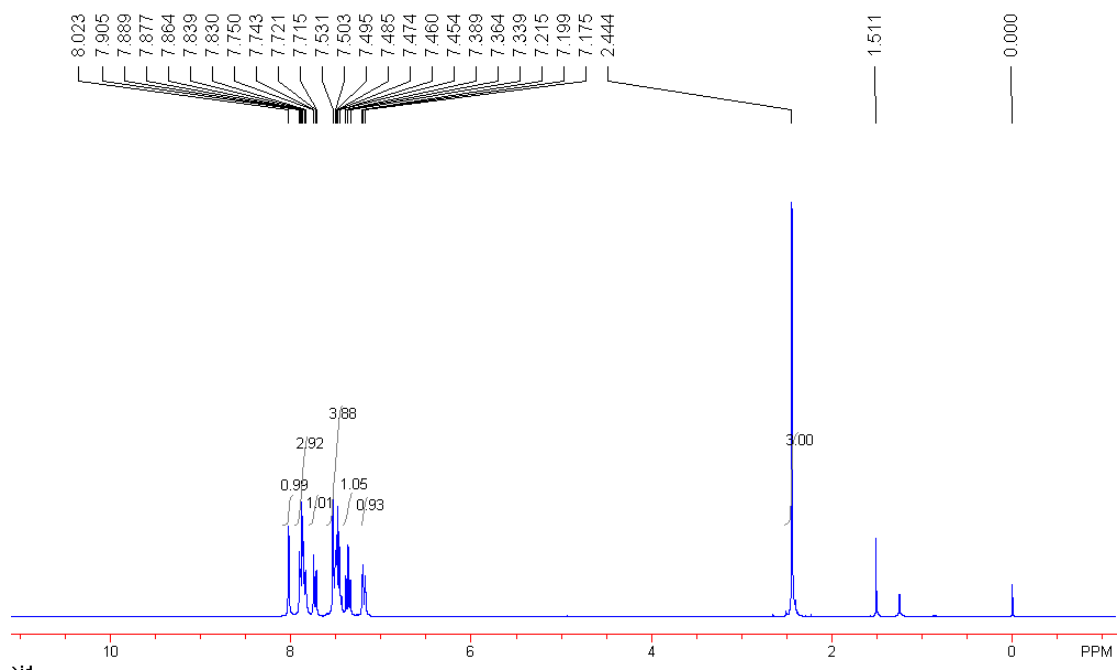
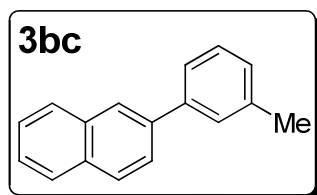
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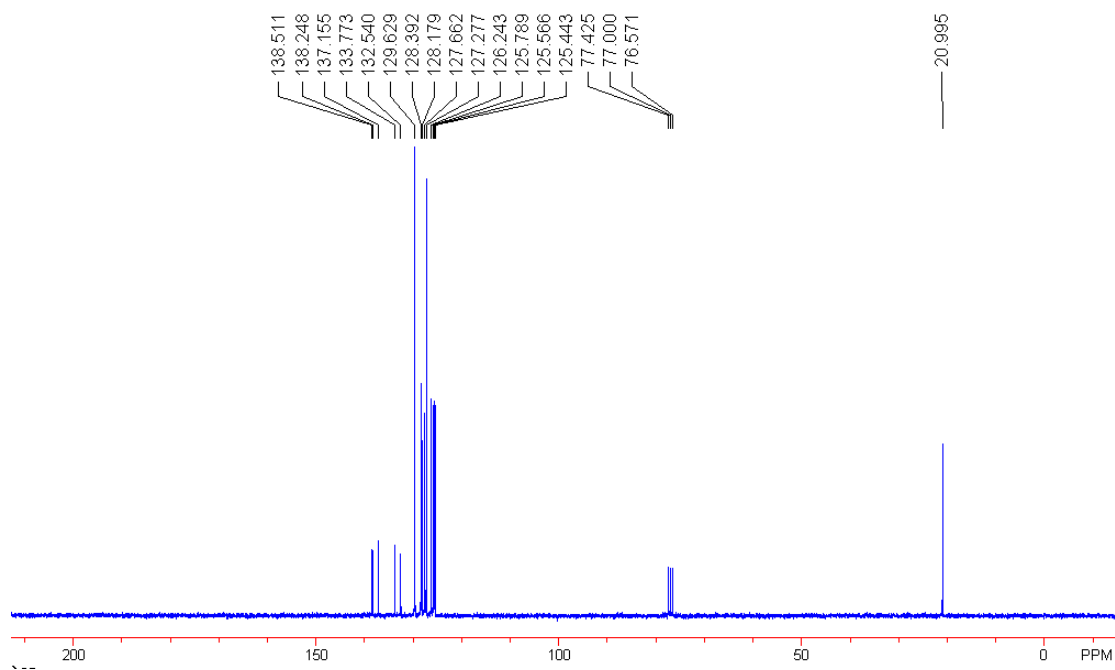
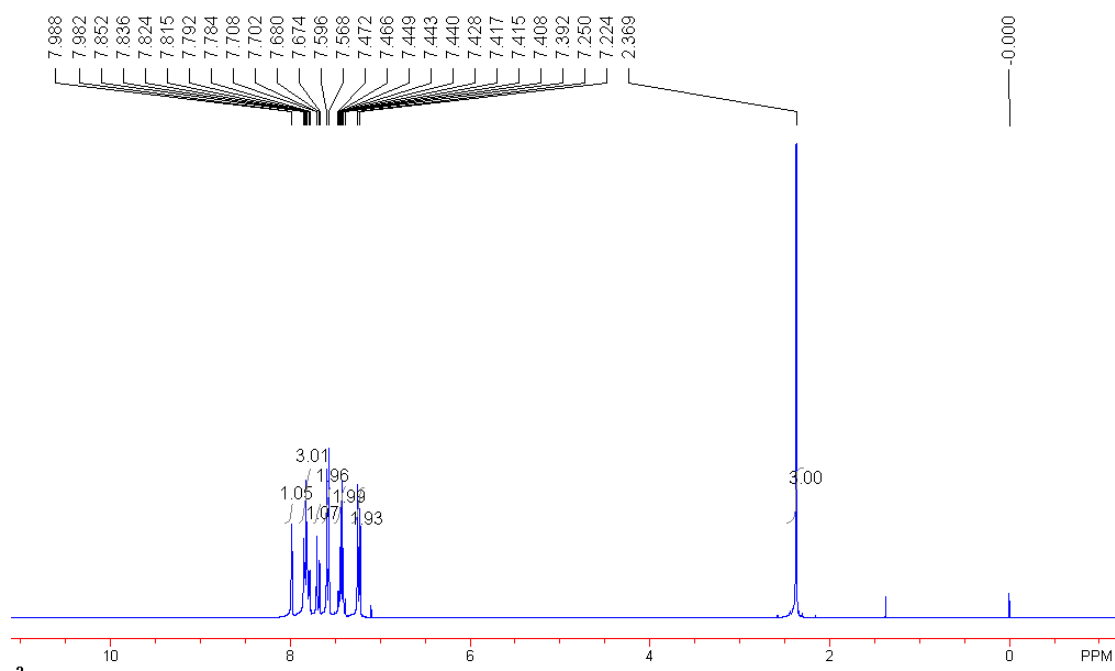
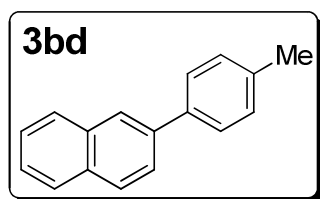
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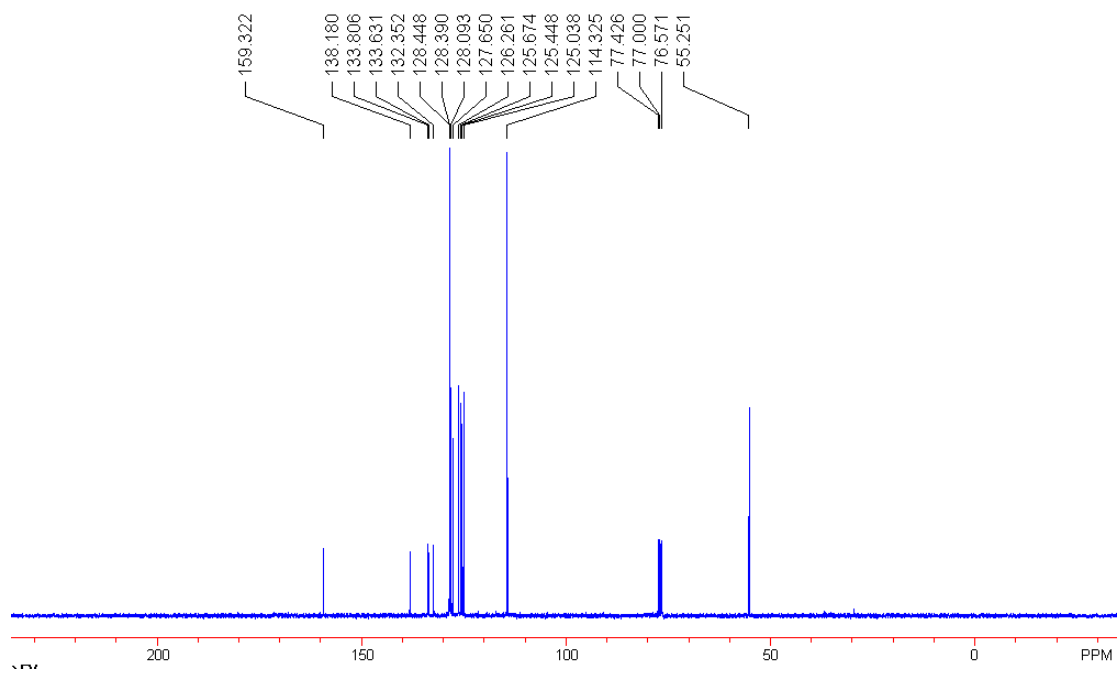
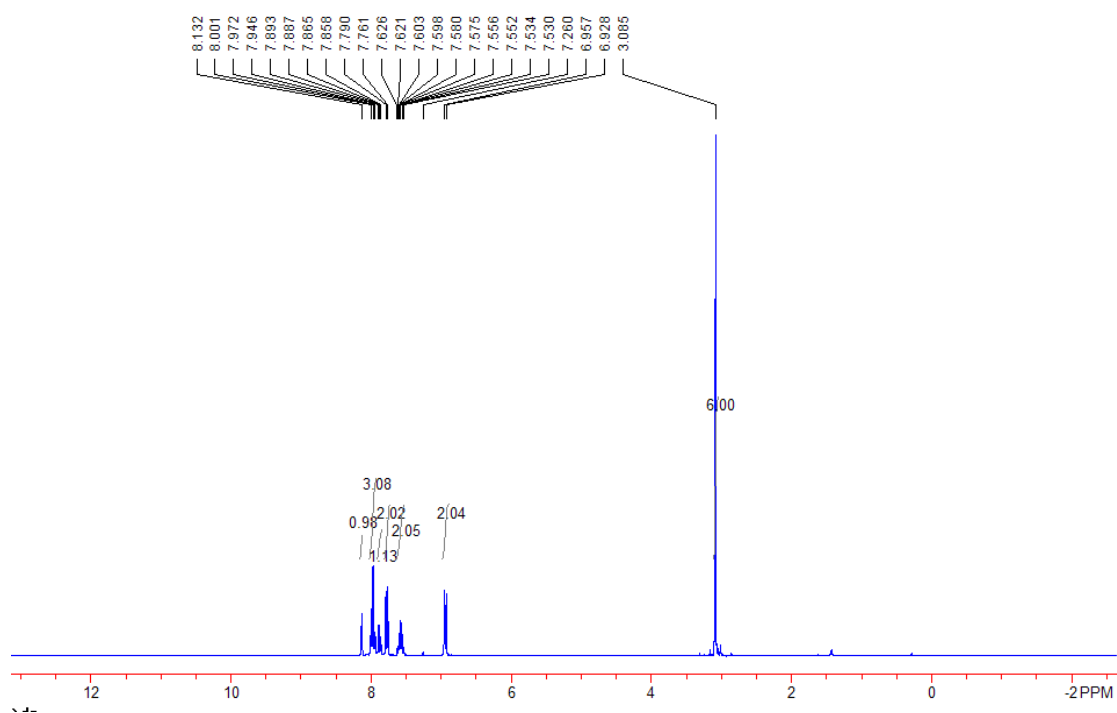
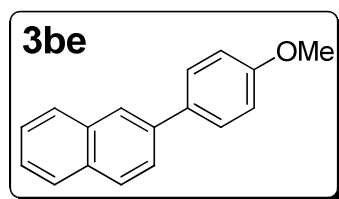


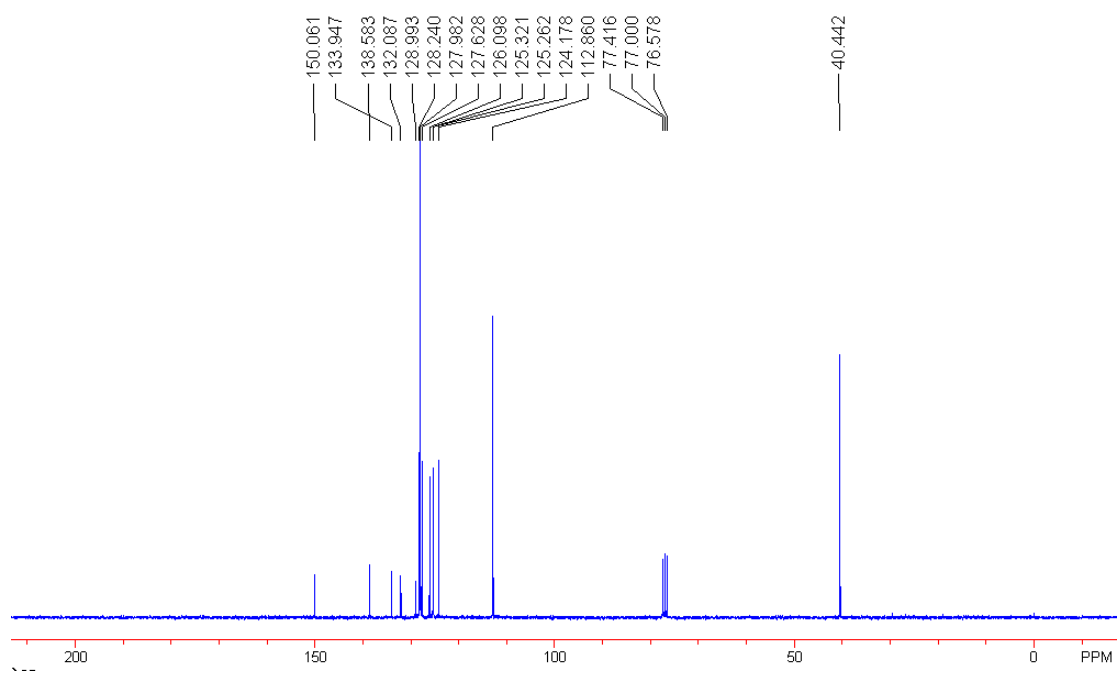
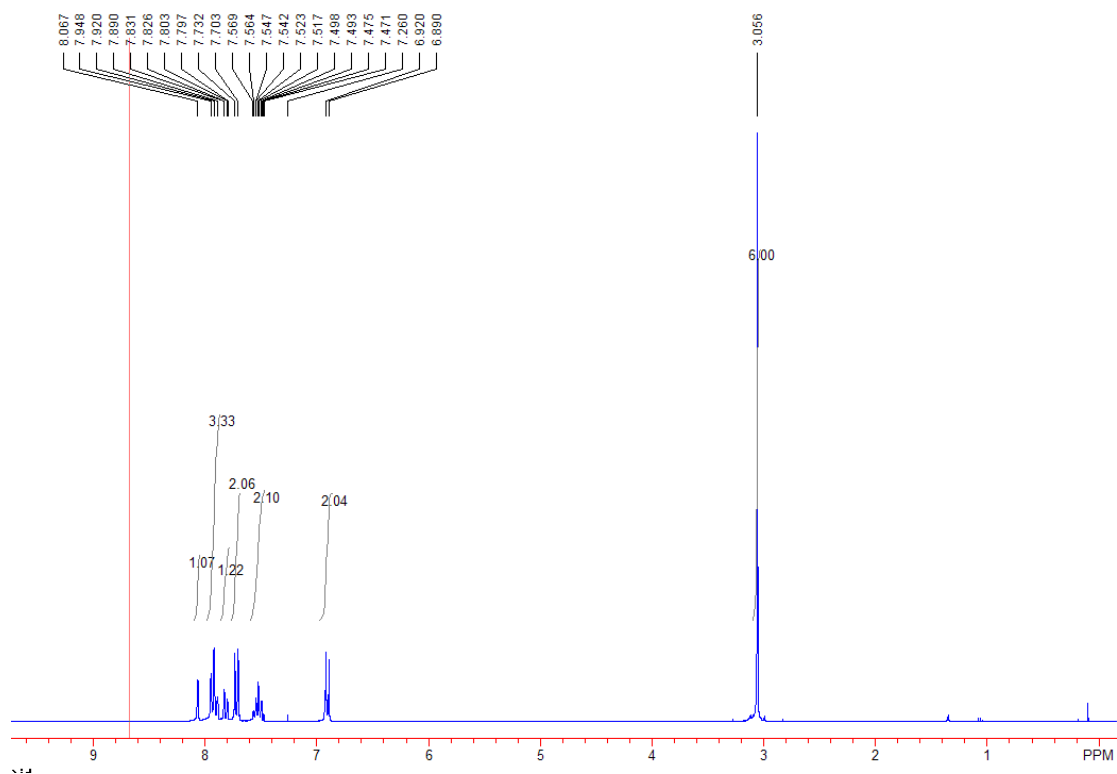
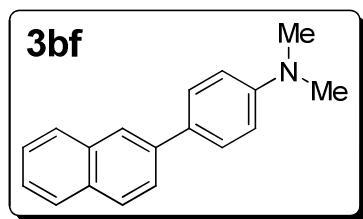


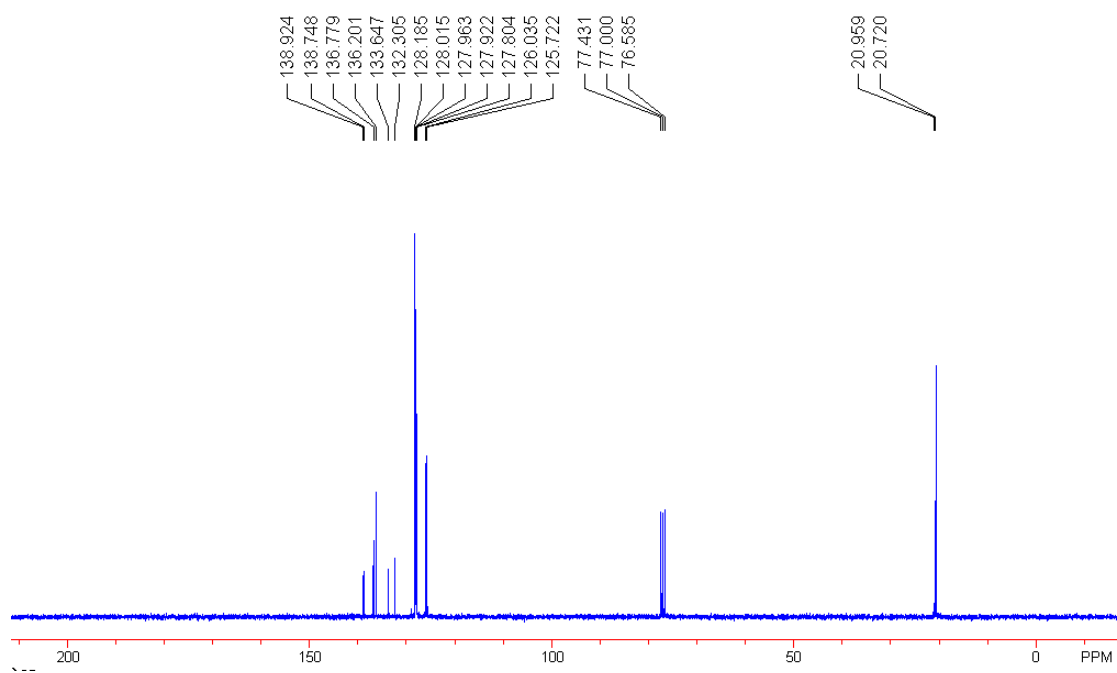
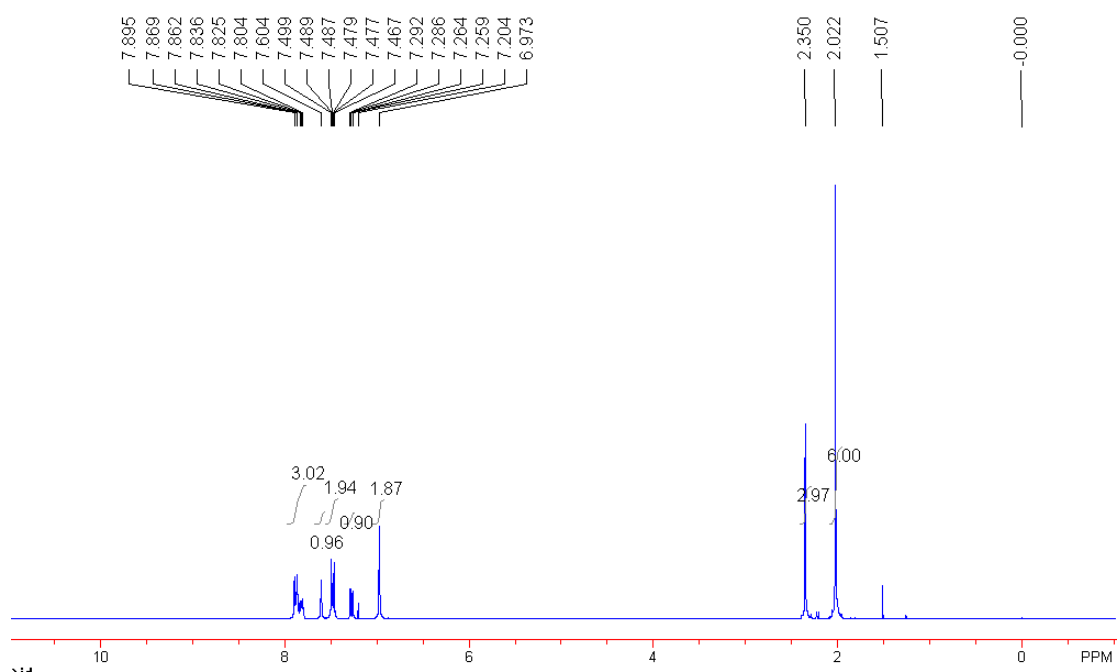
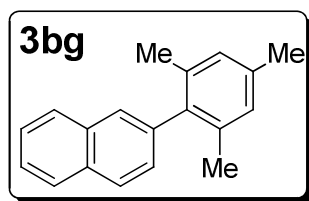


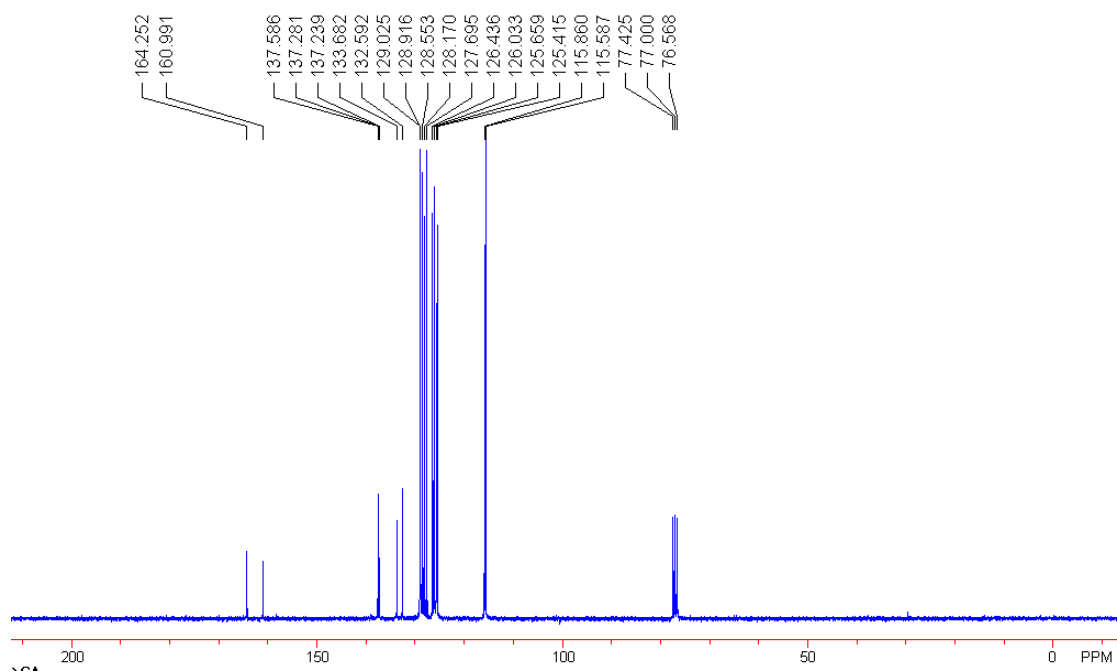
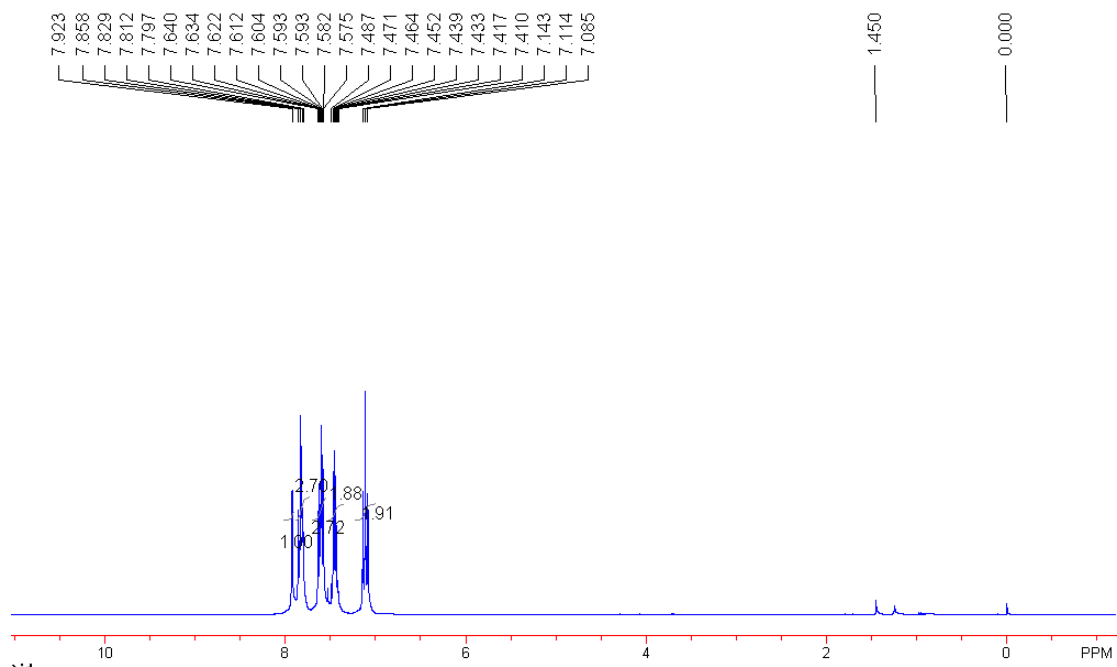
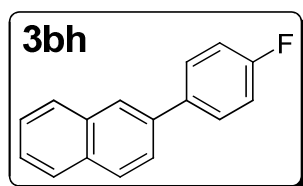


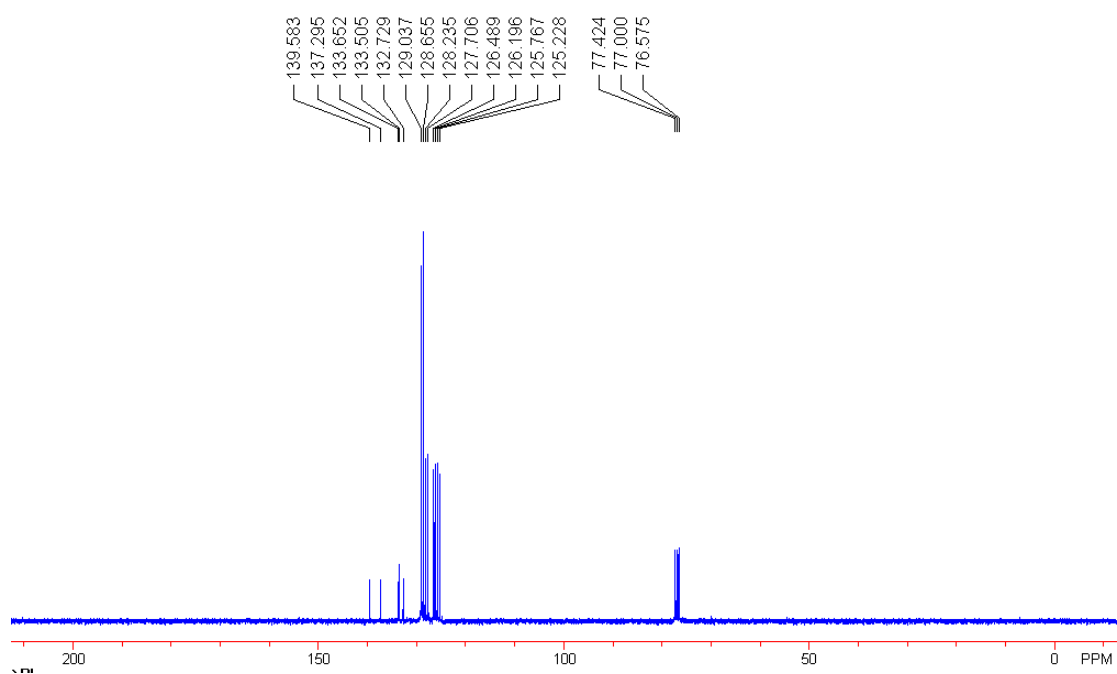
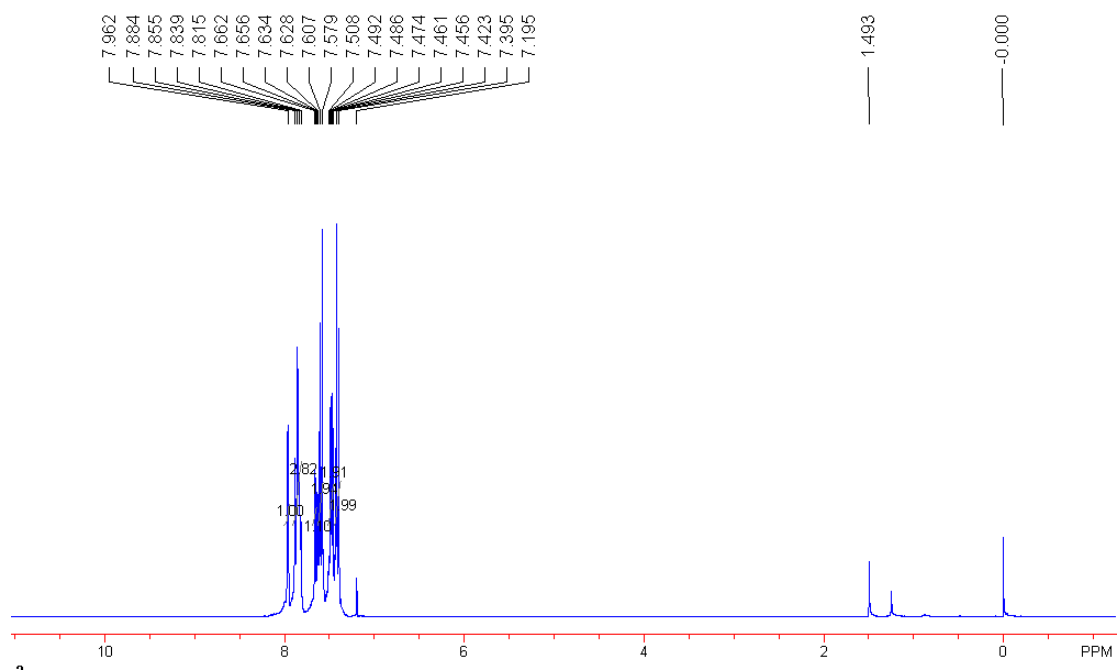


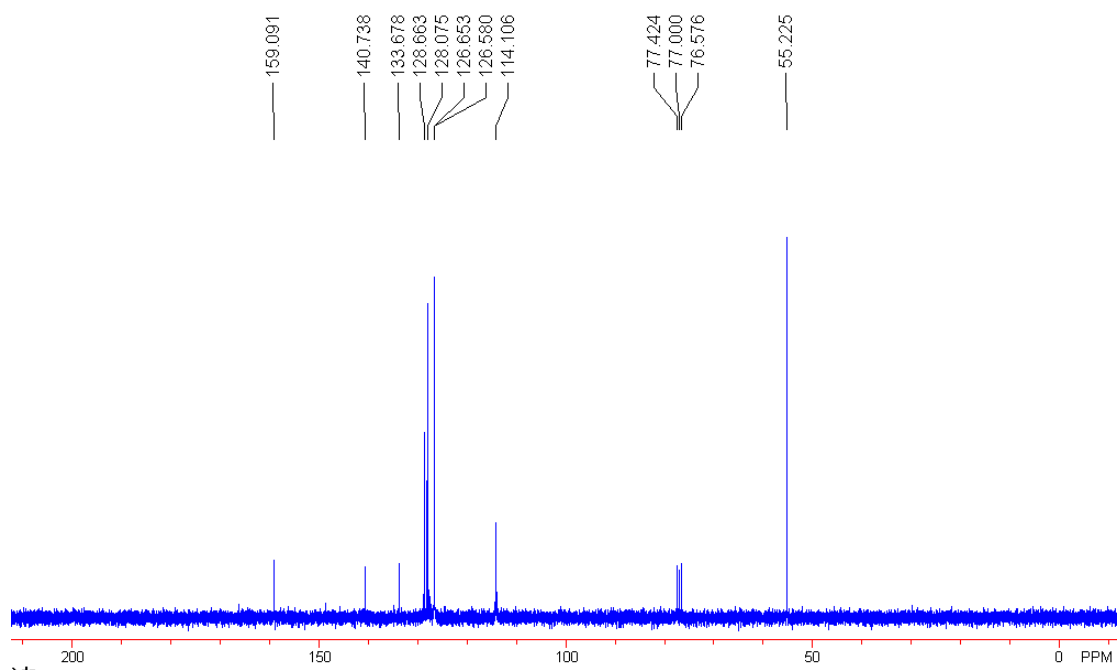
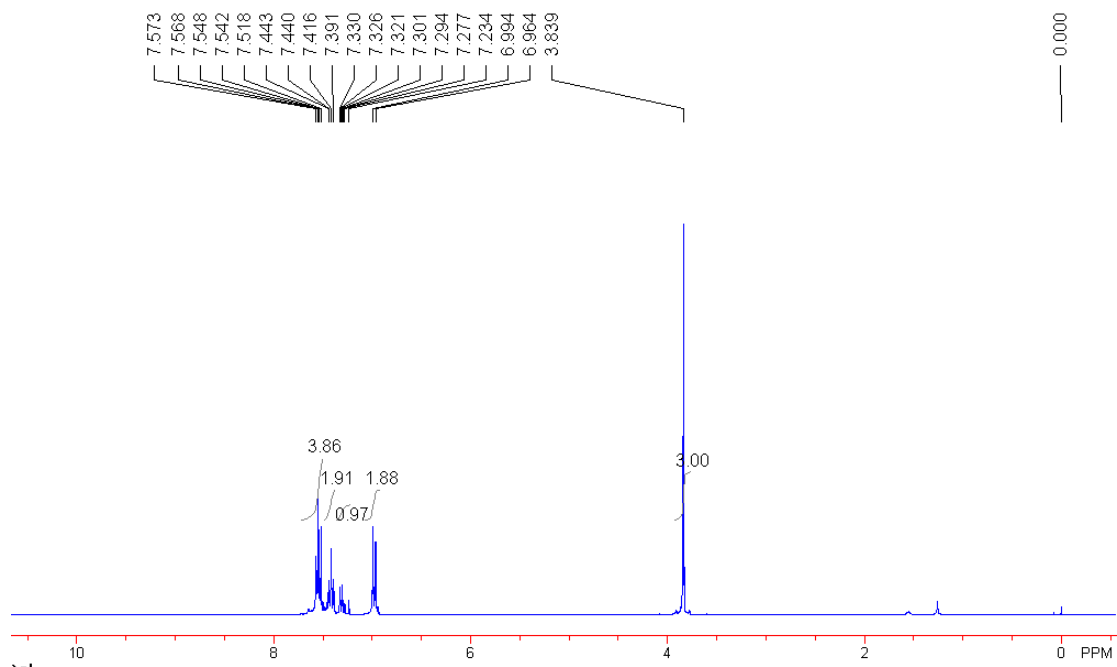
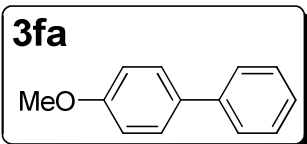




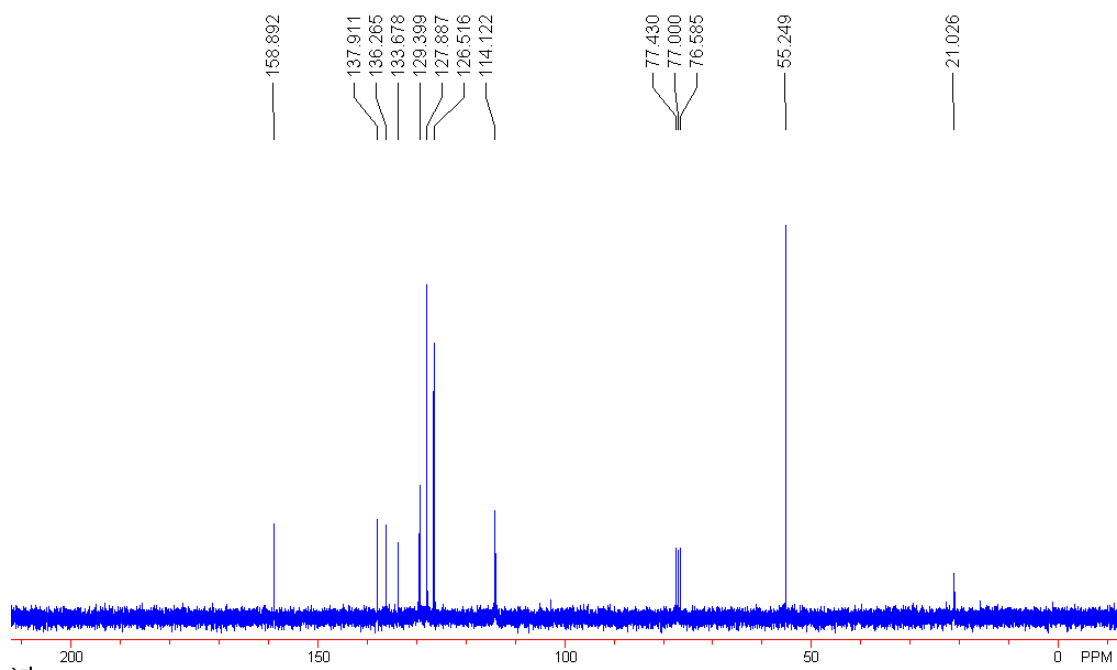
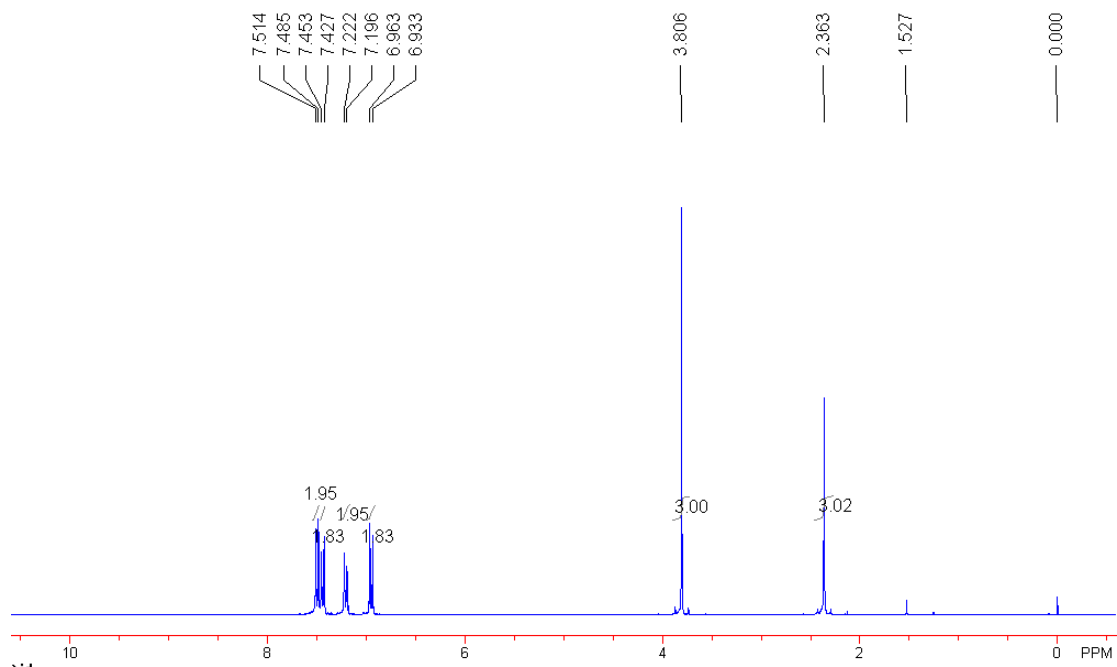
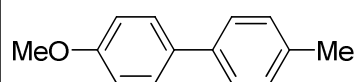


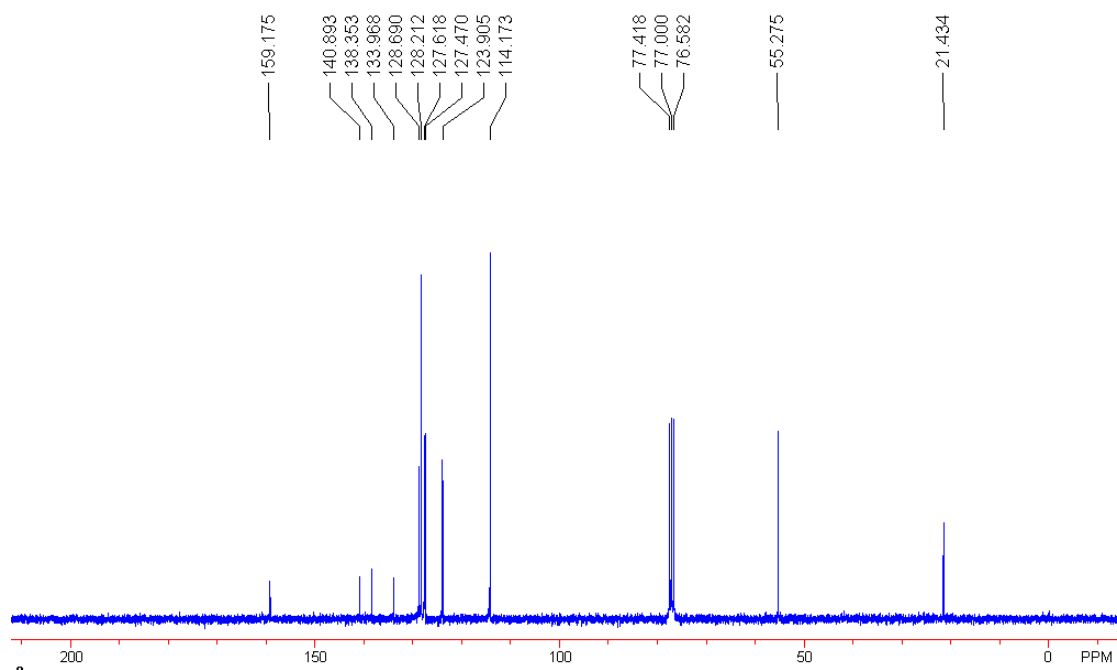
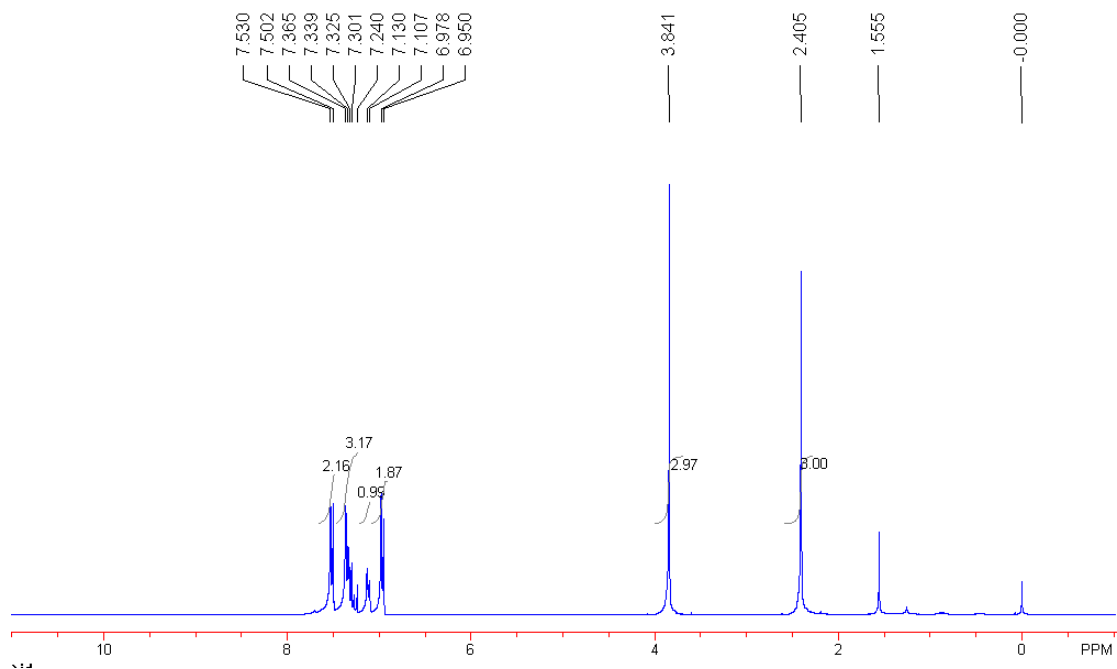
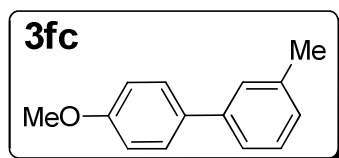


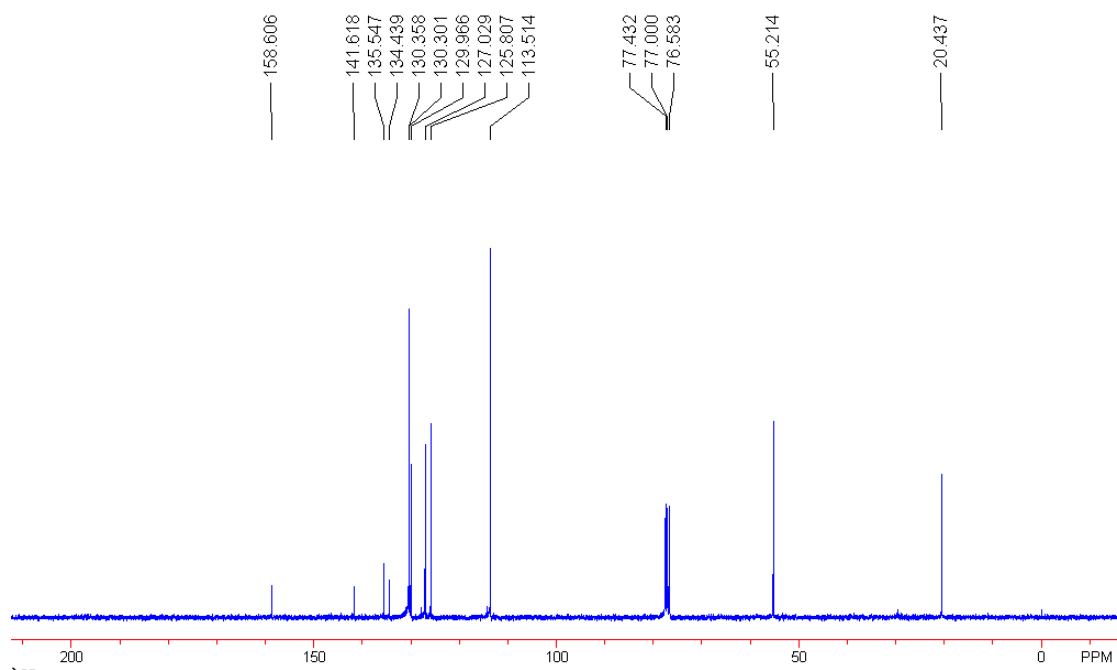
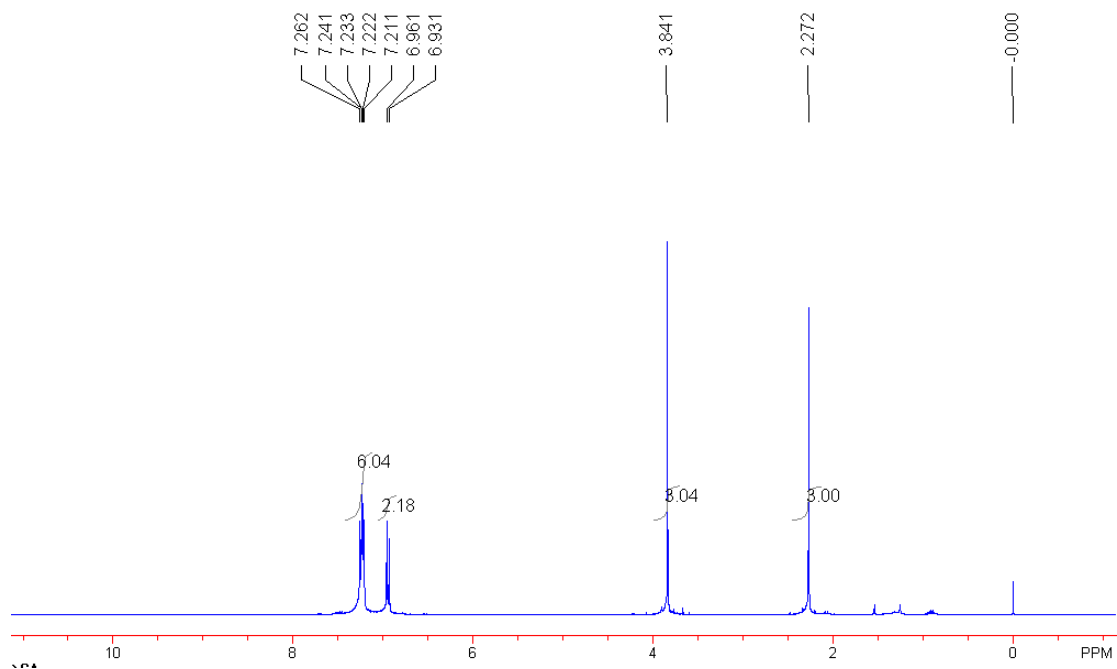
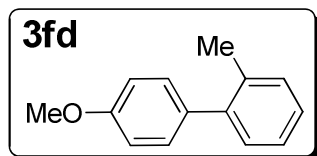


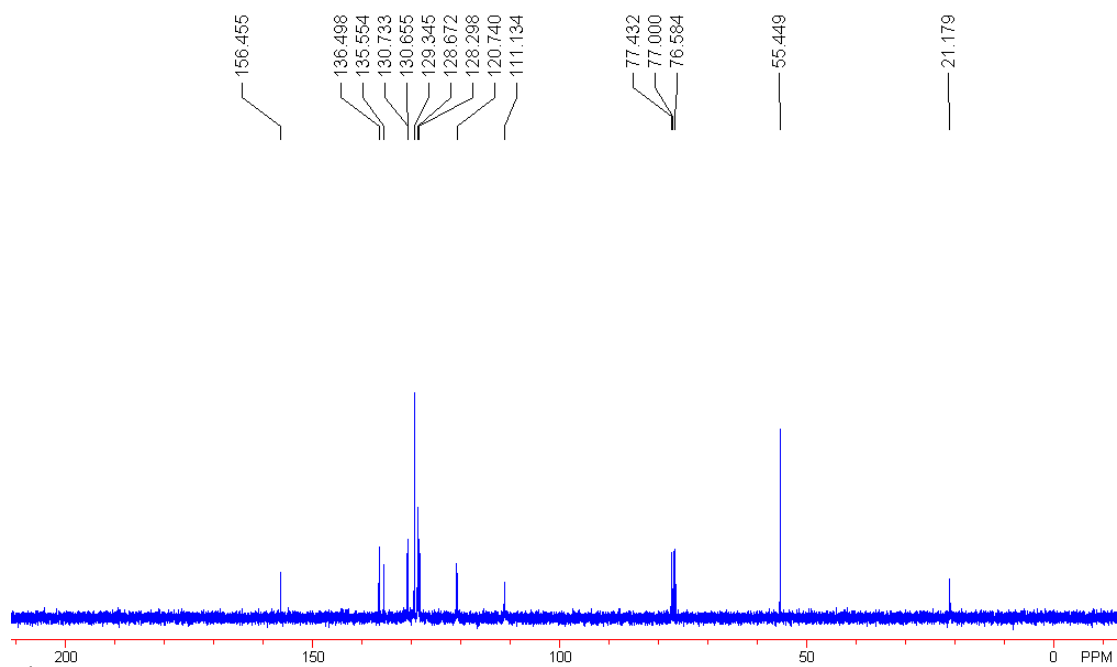
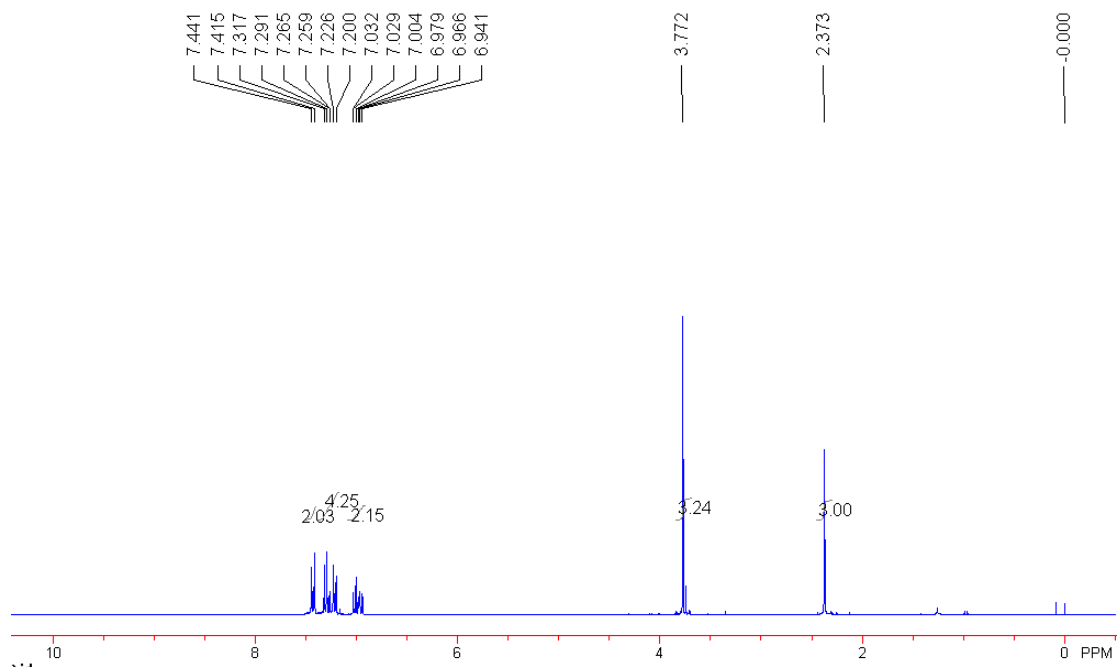
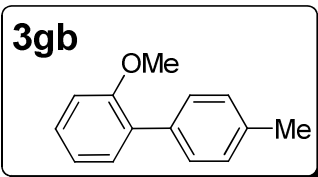


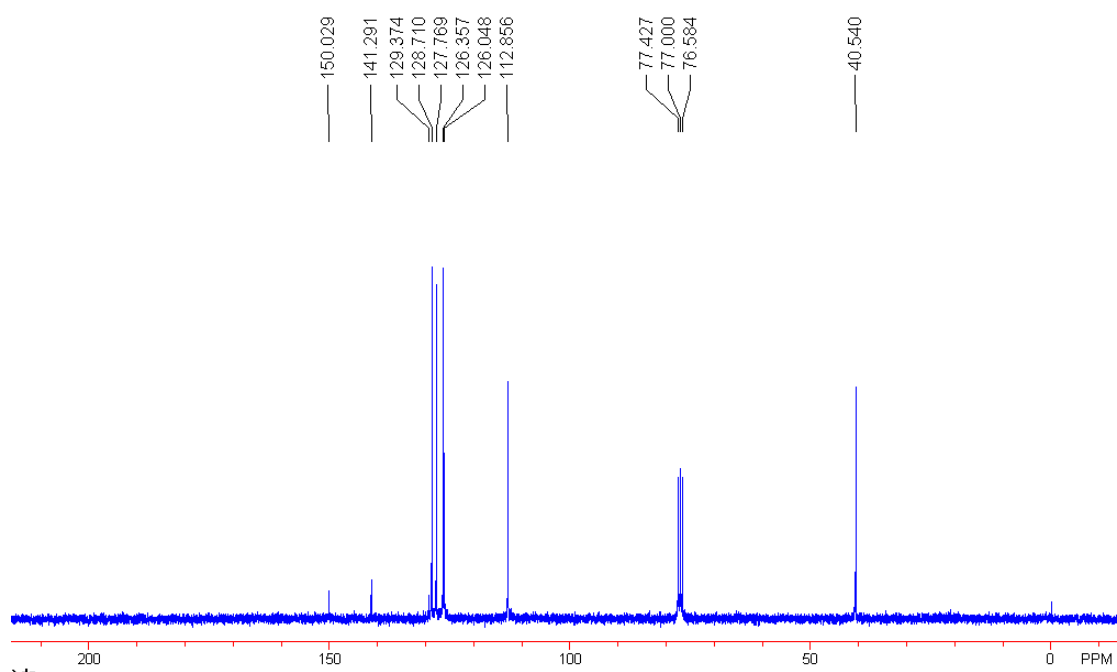
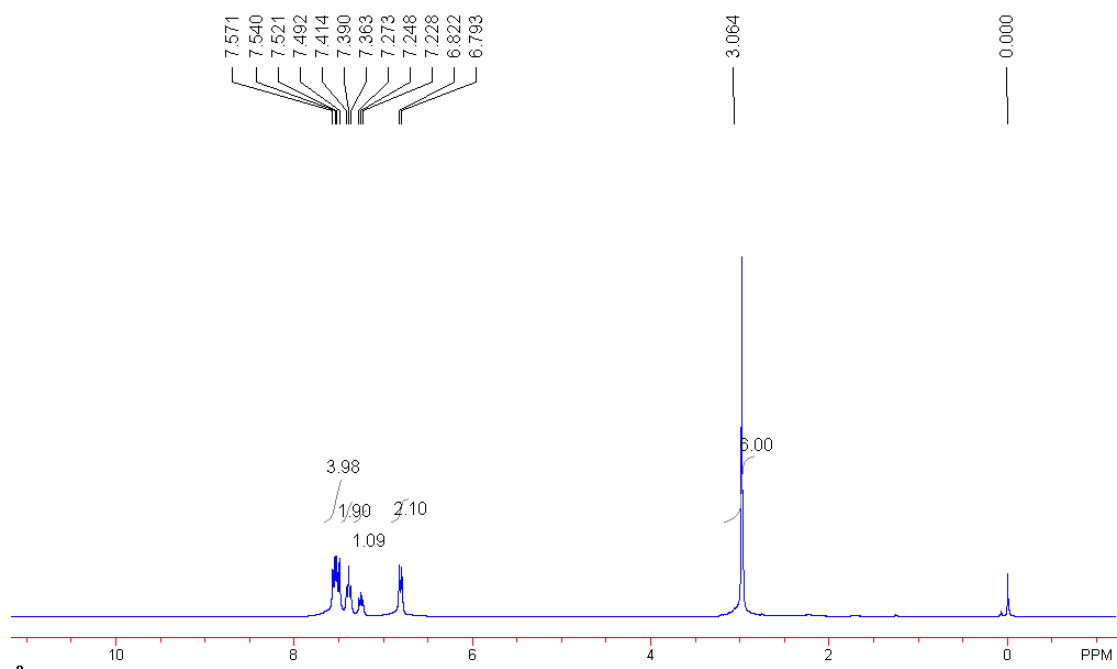
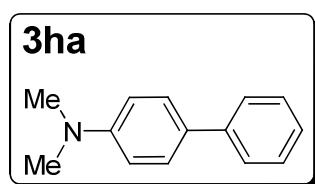
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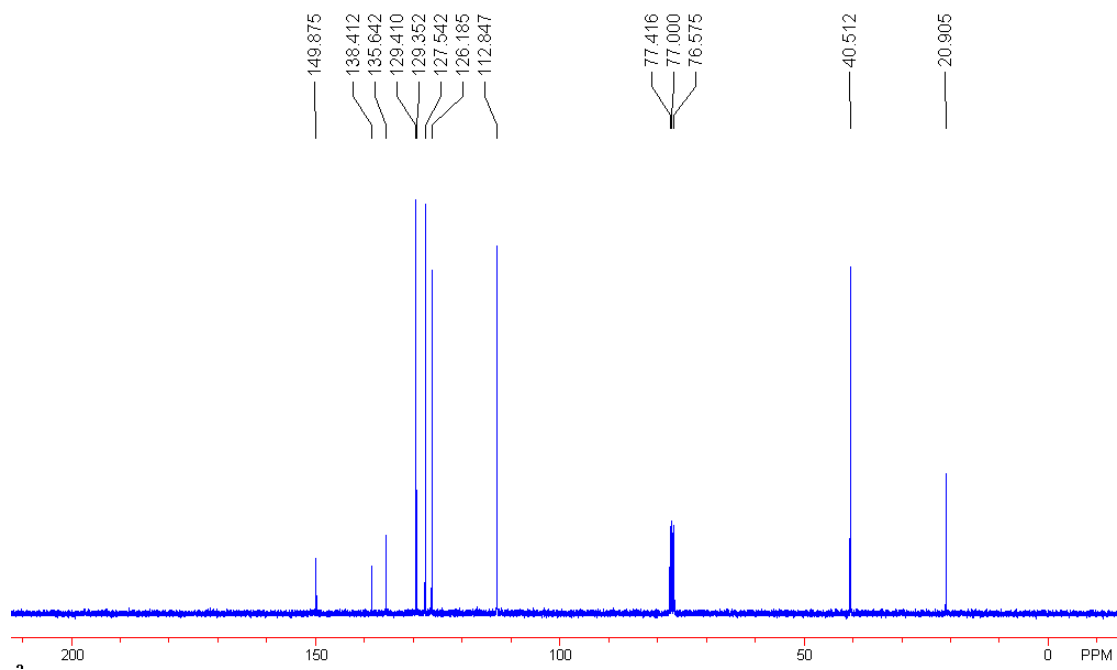
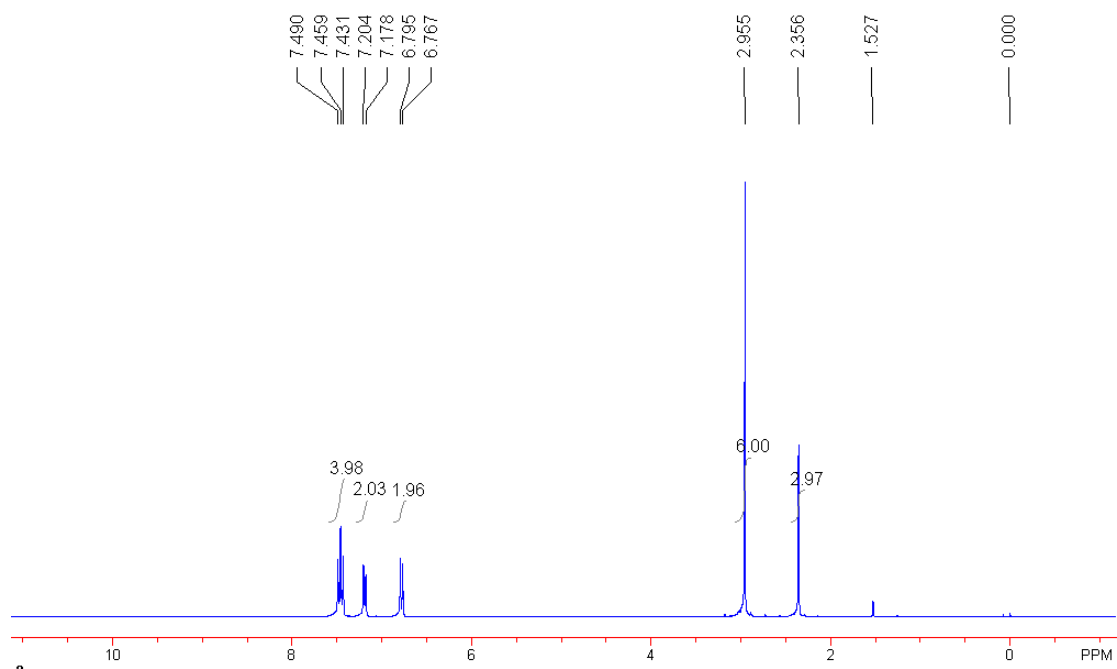
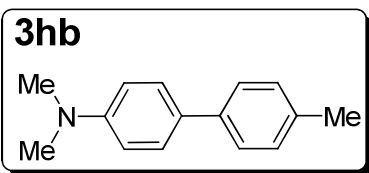


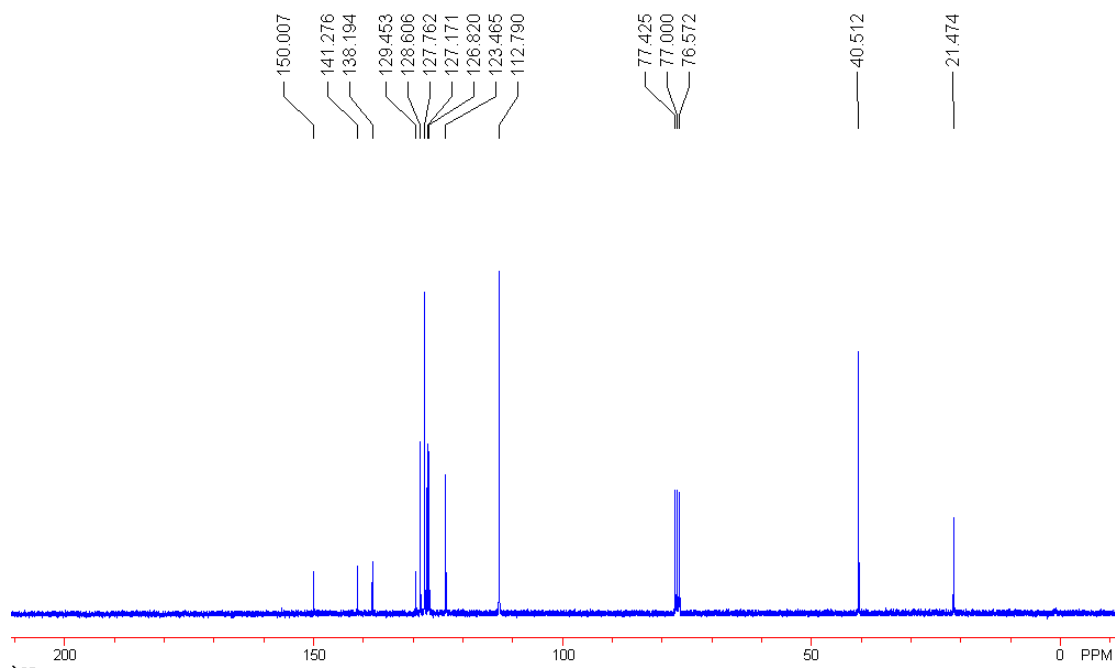
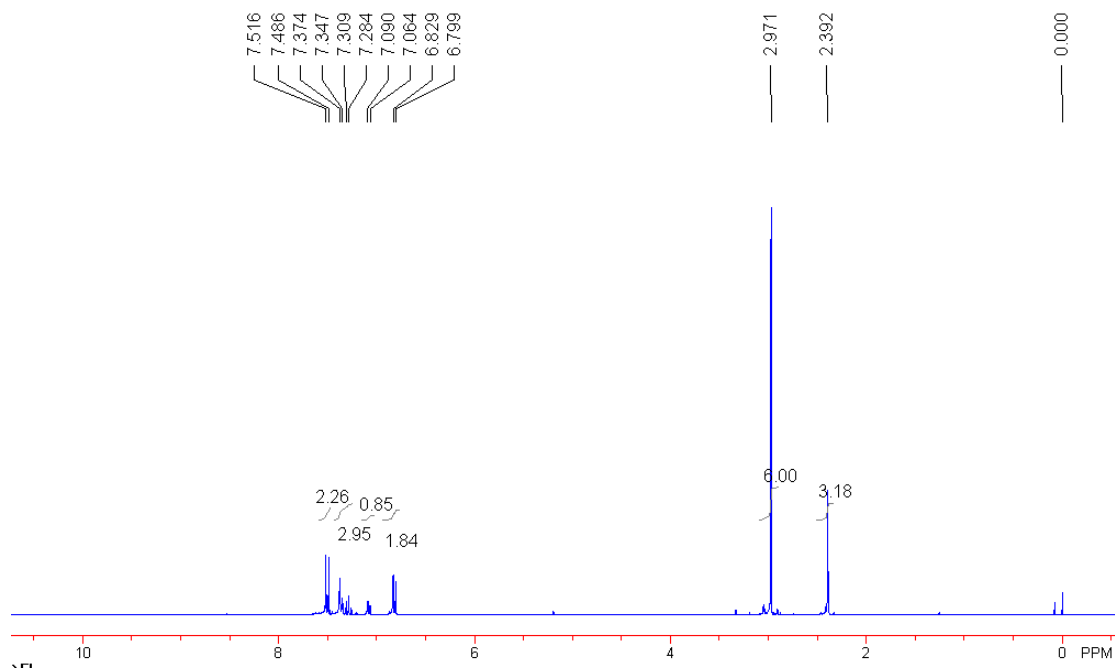
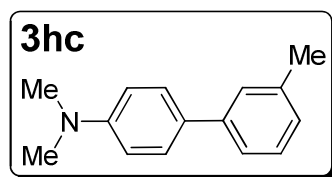


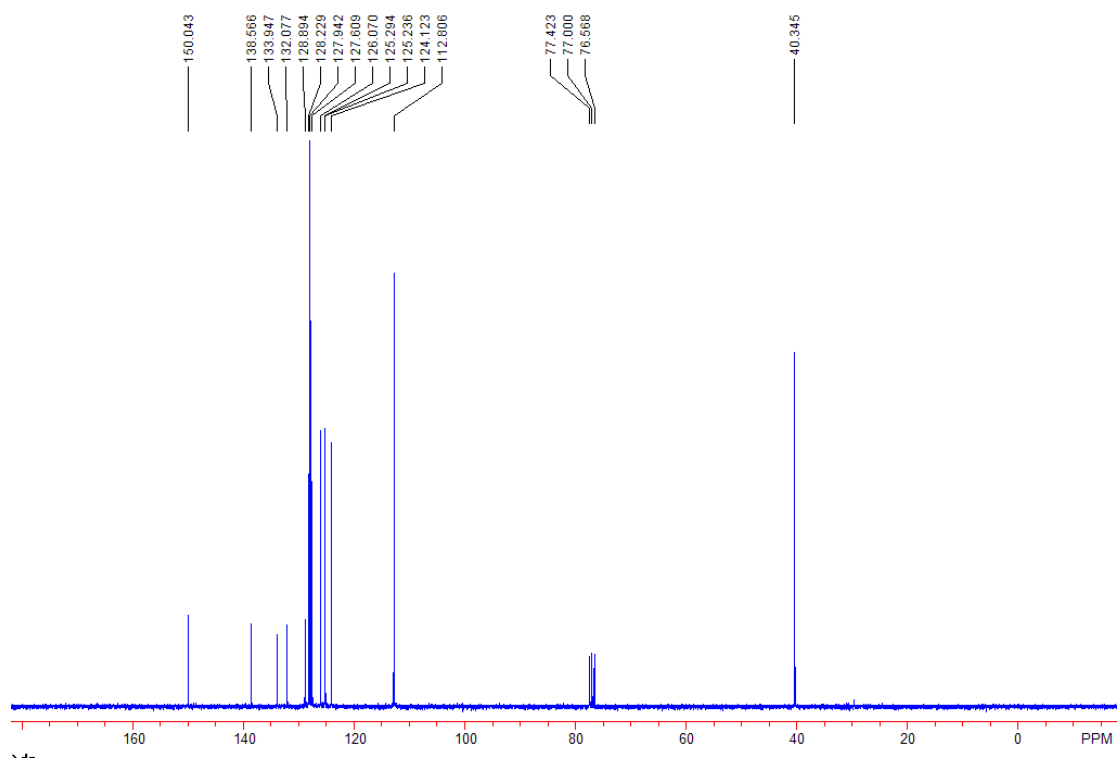
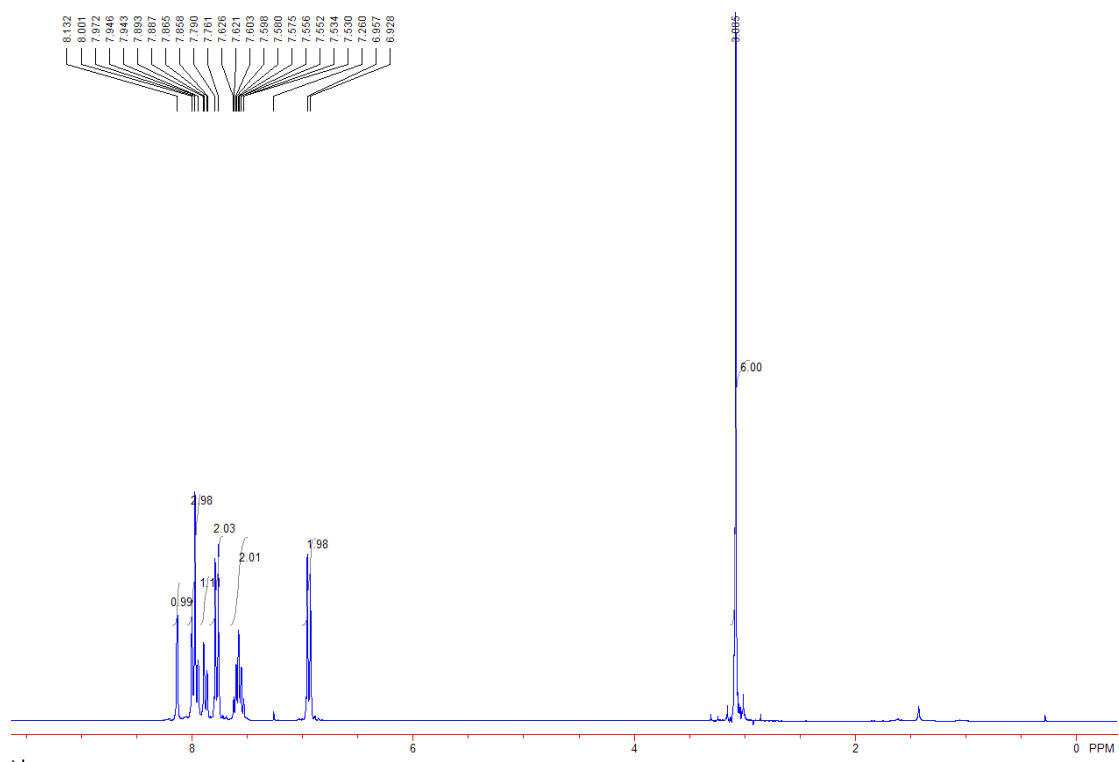
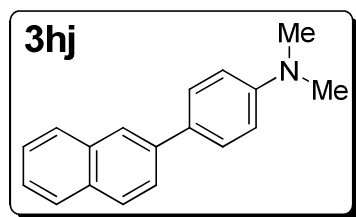


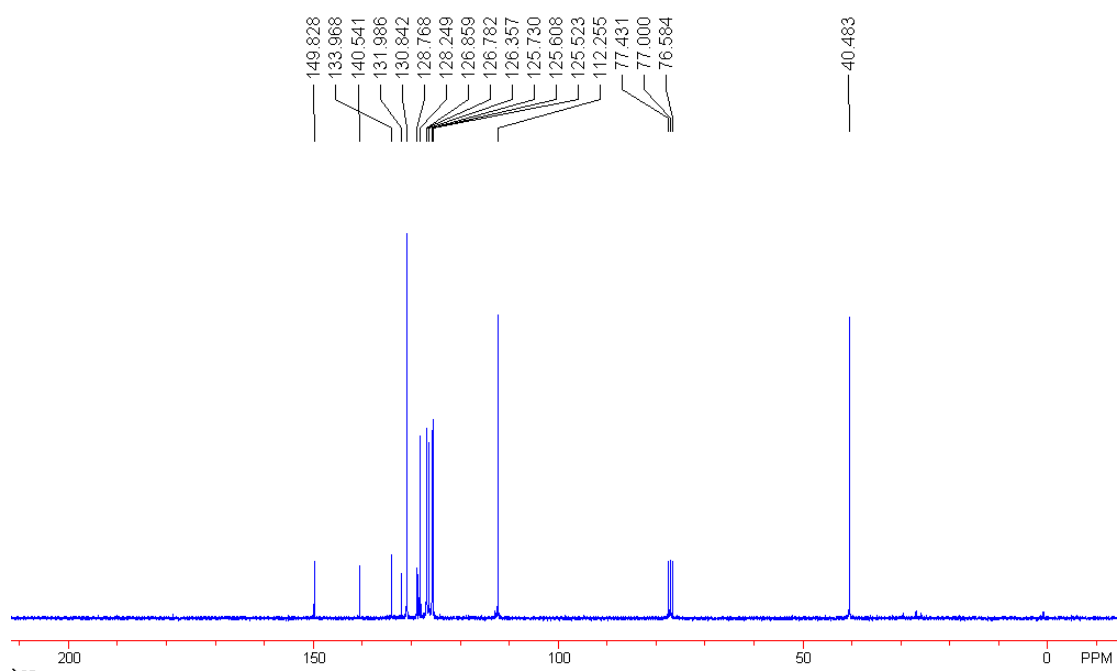
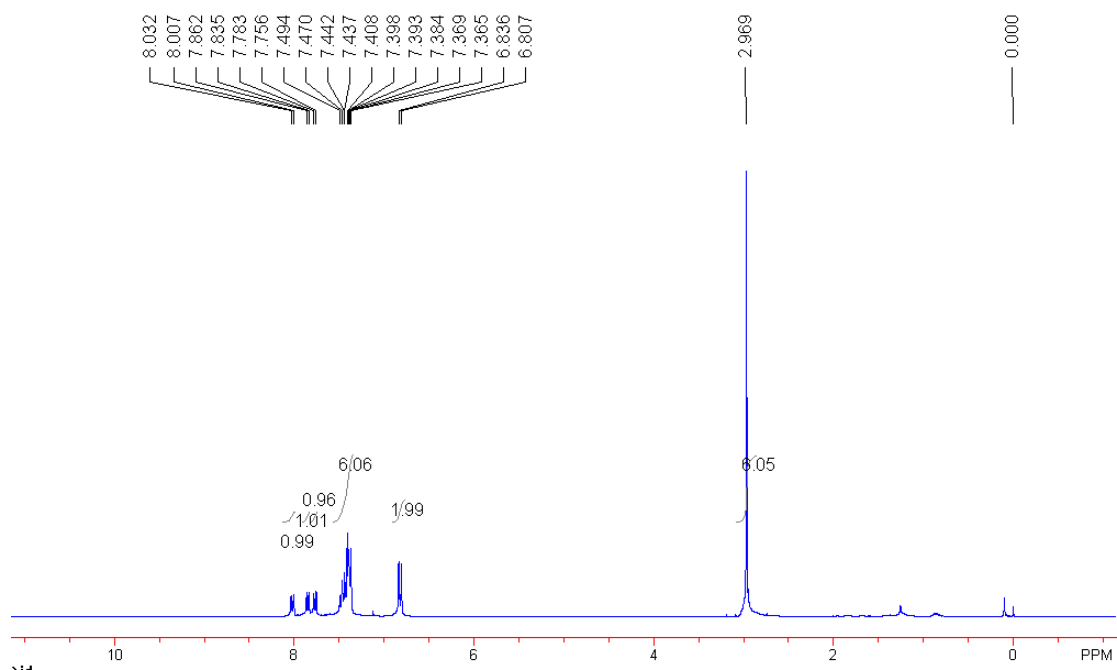
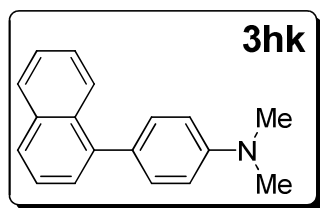




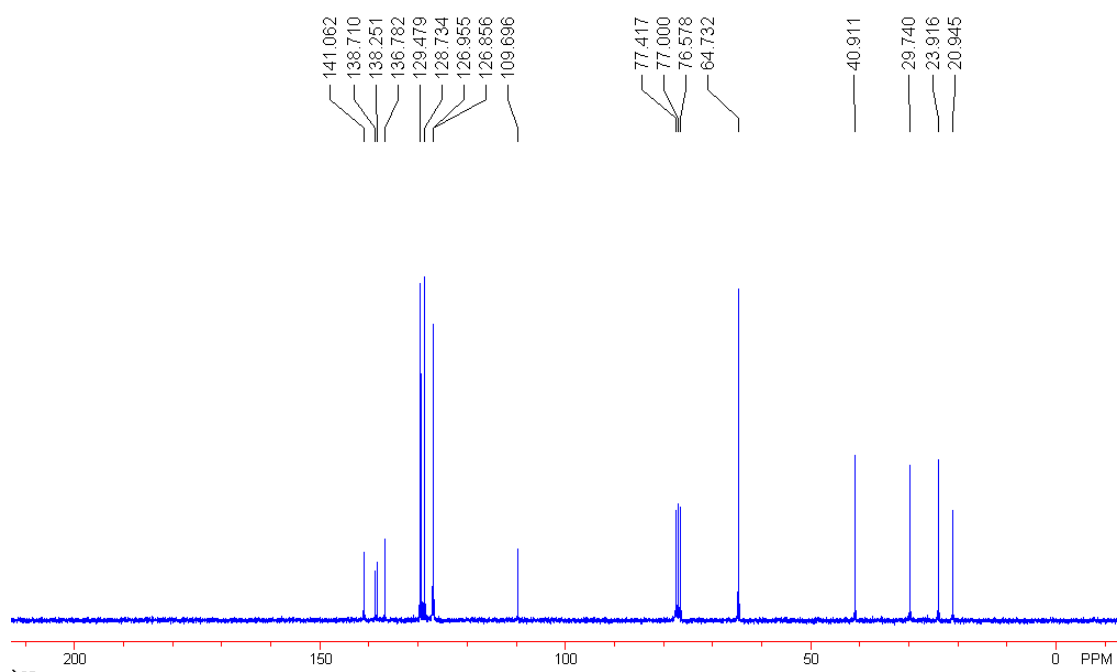
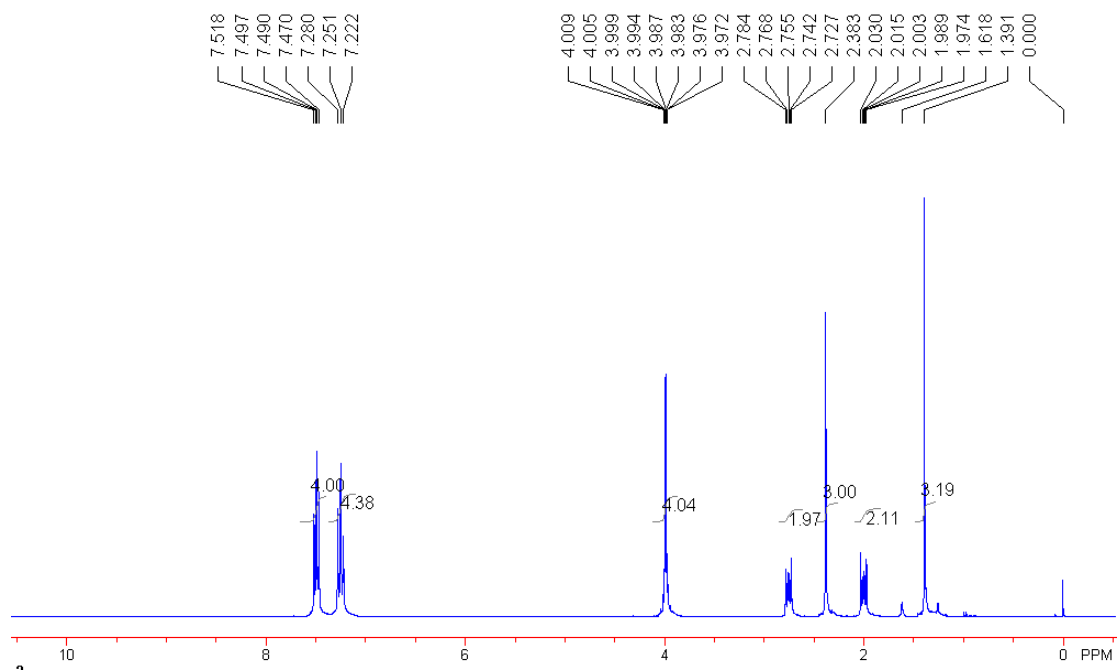
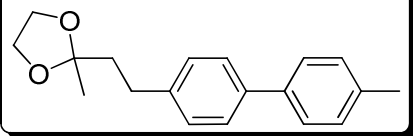


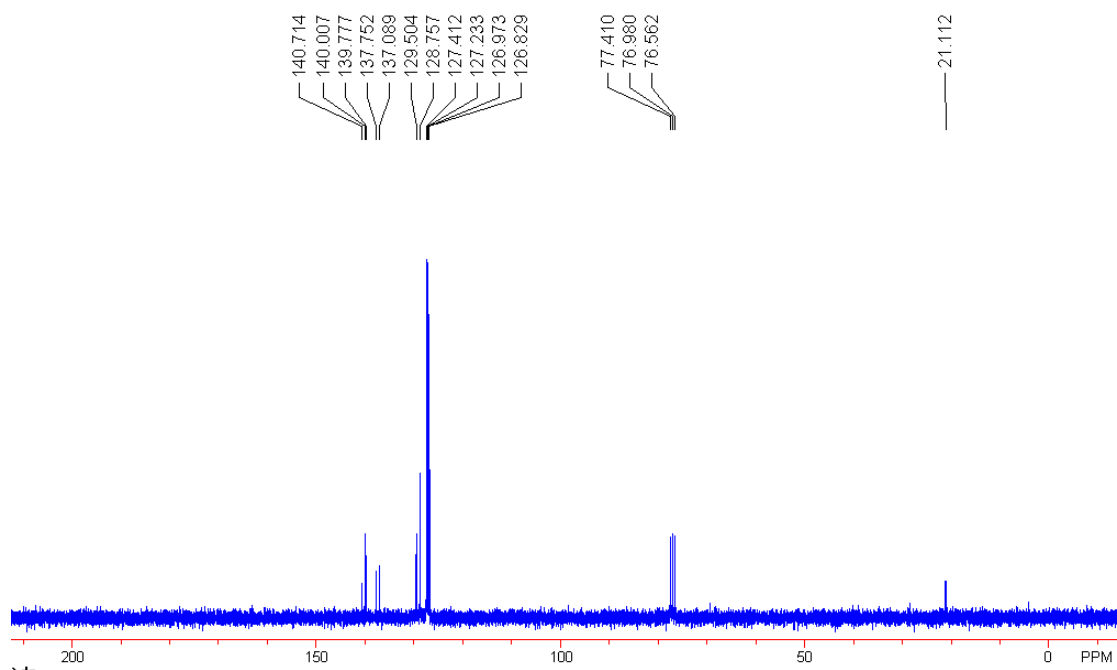
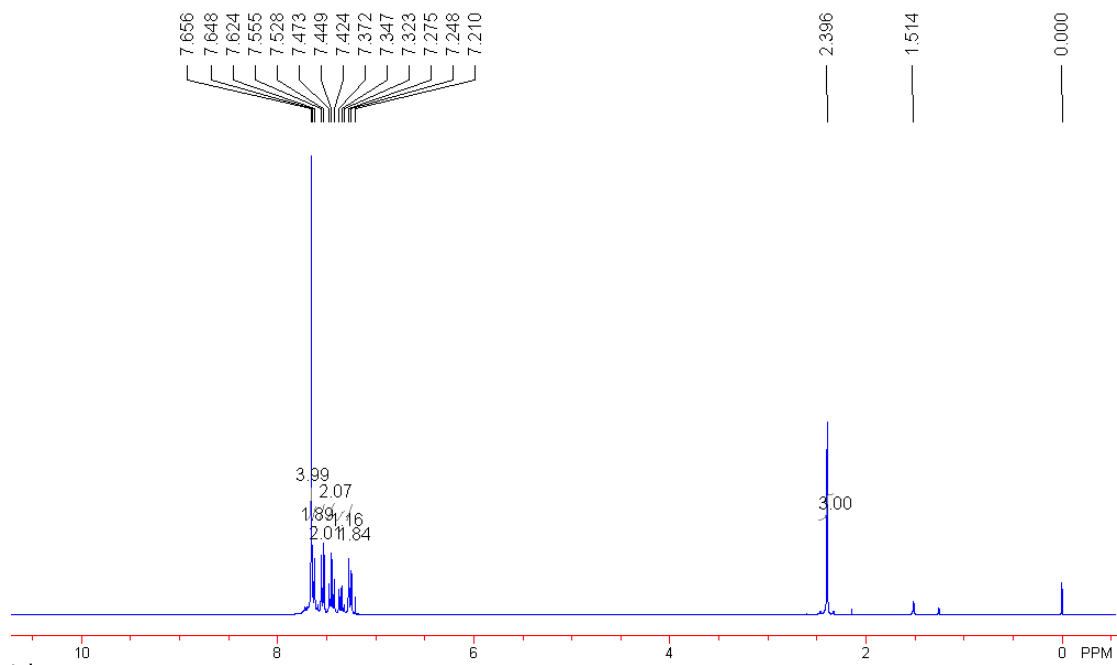
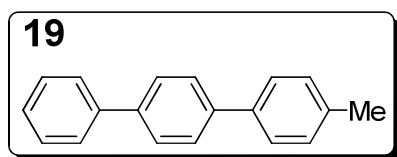




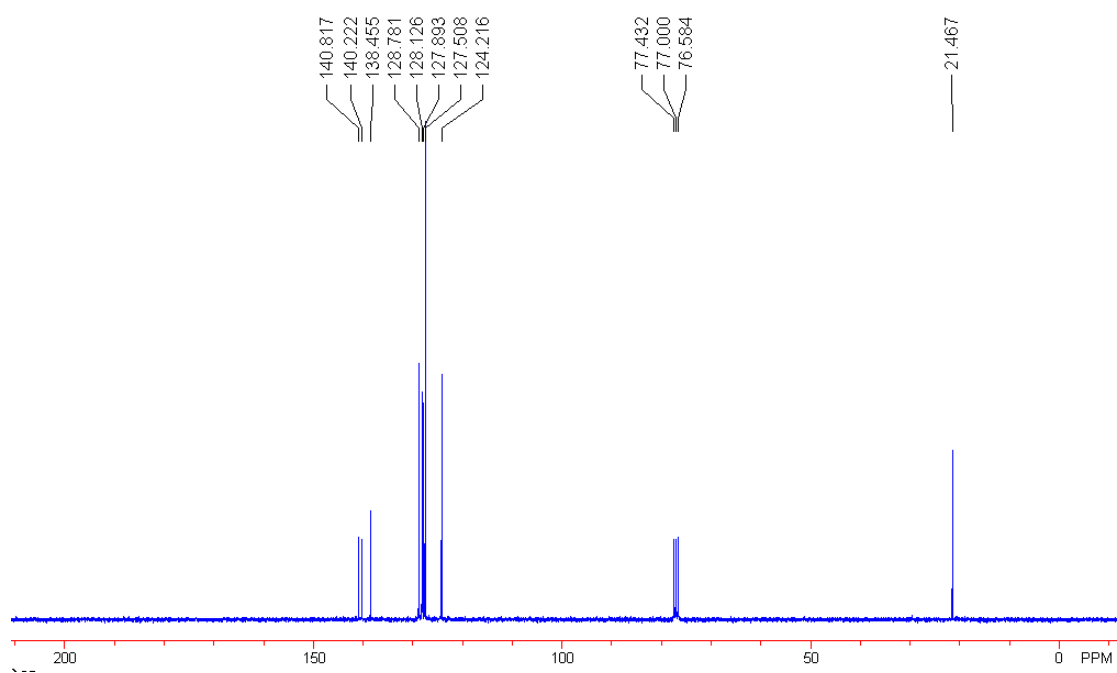
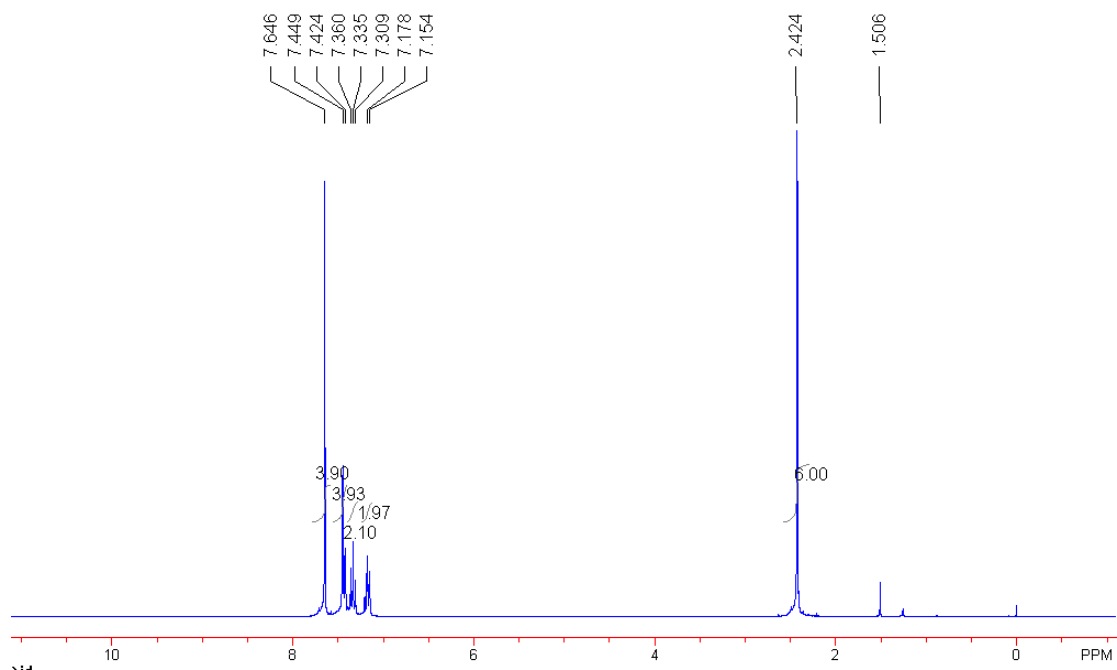
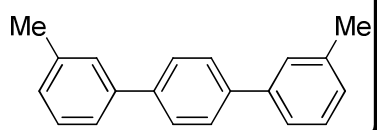


16





Byproduct of 3fc



References:

- [1] M. Tobisu, T. Shimasaki, N. Chatani, *Angew. Chem. Int. Edit.* **2008**, *47*, 4866.
- [2] O. Kobayashi, D. Uraguchi, T. Yamakawa, *Org. Lett.* **2009**, *11*, 2679.
- [3] M. E. Limmert, A. H. Roy, J. F. Hartwig, *J. Org. Chem.* **2005**, *70*, 9364.
- [4] T. K. Macklin, V. Snieckus, *Org. Lett.* **2005**, *7*, 2519.
- [5] C. M. Brito, D. C. G. A. Pinto, A. M. S. Silva, A. M. G. Silva, A. C. Tom, J. A. S. Cavaleiro, *E. J. Org. Chem.* **2006**, *2006*, 2558.
- [6] J. Albaneze-Walker, R. Raju, J. A. Vance, A. J. Goodman, M. R. Reeder, J. Liao, M. T. Maust, P. A. Irish, P. Espino, D. R. Andrews, *Org. Lett.* **2009**, *11*, 1463.
- [7] J. Liu, Y. Deng, H. Wang, H. Zhang, G. Yu, B. Wu, H. Zhang, Q. Li, T. B. Marder, Z. Yang, A. Lei, *Org. Lett.* **2008**, *10*, 2661.
- [8] A. H. Roy, J. F. Hartwig, *J. Am. Chem. Soc.* **2003**, *125*, 8704.
- [9] N. Yoshikai, H. Mashima, E. Nakamura, *J. Am. Chem. Soc.* **2005**, *127*, 17978.
- [10] K. Inada, N. Miyaoura, *Tetrahedron* **2000**, *56*, 8657.
- [11] C. Desmarets, R. Omar-Amrani, A. Walcarius, J. Lambert, B. Champagne, Y. Fort, R. Schneider, *Tetrahedron* **2008**, *64*, 372.
- [12] C. Cho, M. Sun, Y. Seo, C. Kim, K. Park, *J. Org. Chem.* **2005**, *70*, 1482.